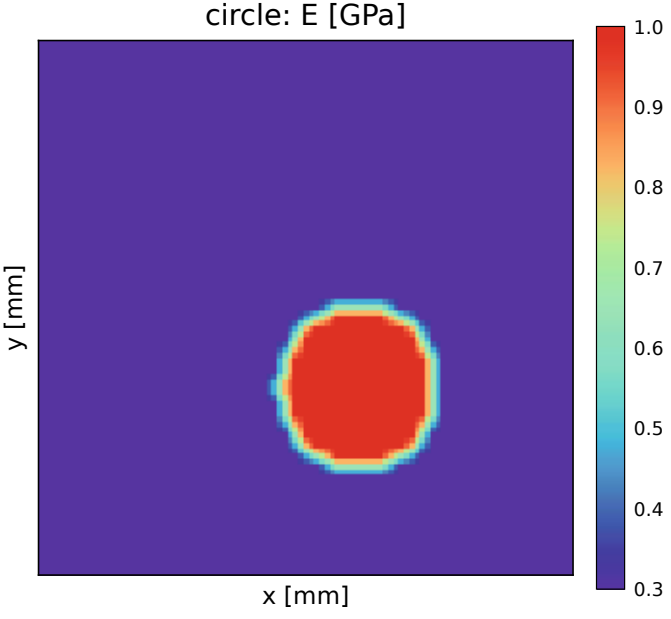
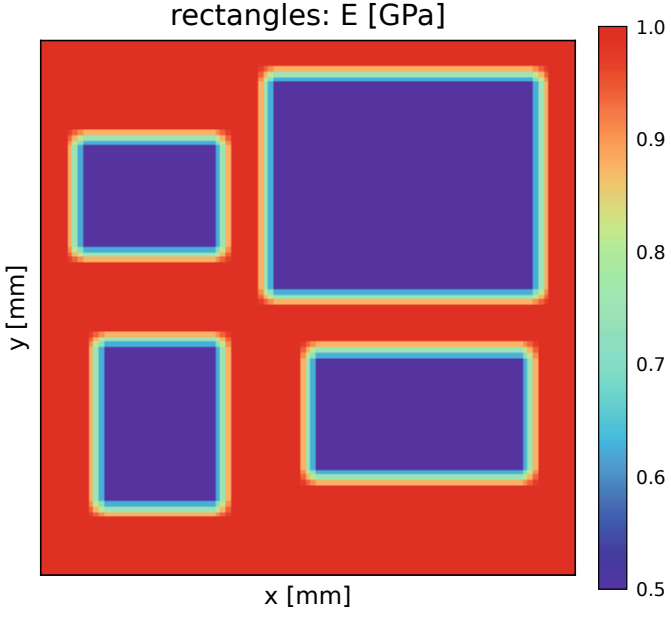




PARTIAL PROGRESS: completed jobs only | Recomputed from shared geometry code; alpha=1,T0=0 and origins are reconstruction choices. Paper unit-label conflict disclosed.

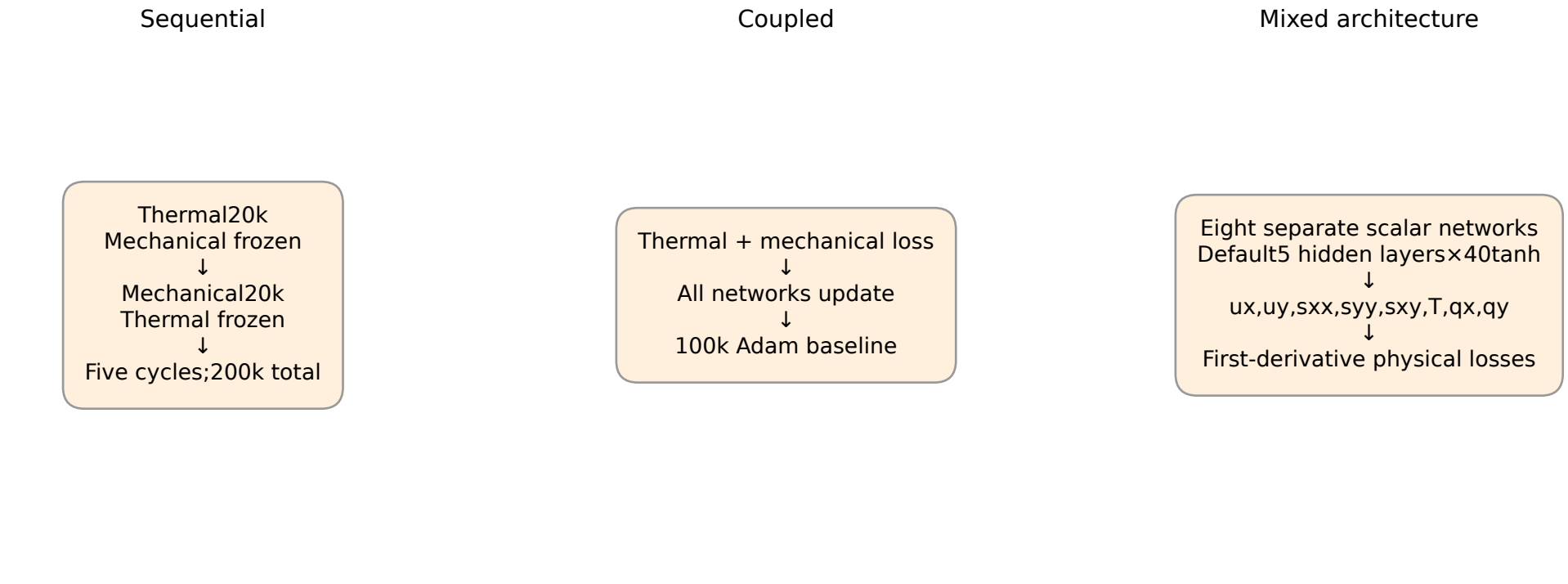
Mechanical: ux=0 at x=0,1; uy=0 at y=0,1

Thermal: T=1 left, T=0 right; qy=0 top and bottom

Soft baseline: boundary penalties in the loss

Hard cases: $\hat{u}_x = x(x-1)u_x$, $\hat{u}_y = y(y-1)u_y$

$\hat{T} = x(x-1)T + 1 - x$



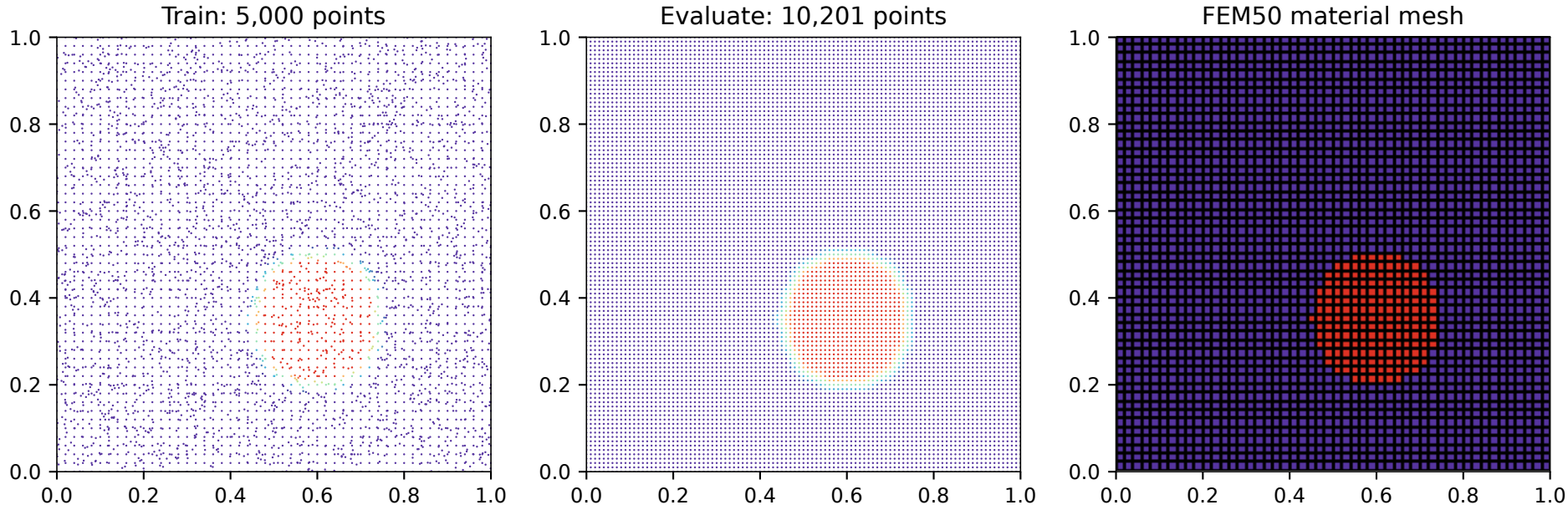
PARTIAL PROGRESS: completed jobs only | New schematic. Baseline soft BCs and authors-energy variant; hard/L-BFGS and transfer cases have separate configurations.

Paired material ratios (1,1) → … → (10,10)

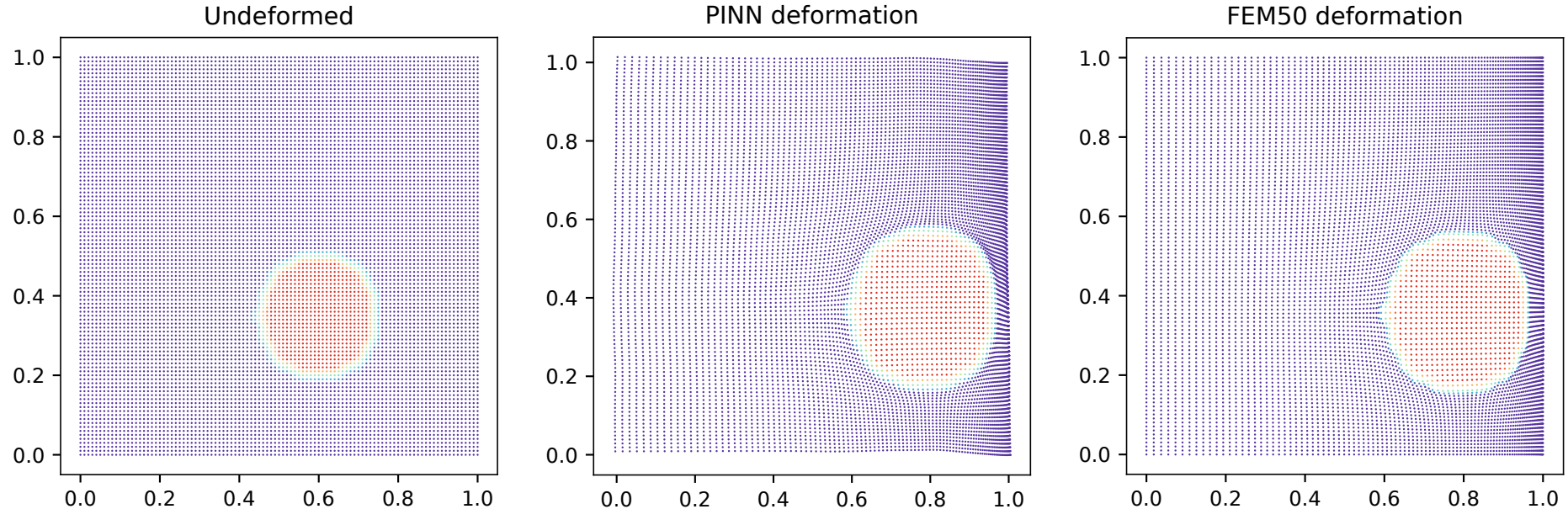
10,000 Adam updates per ratio; transfer network weights

Evaluate held-out (15,15) without further training

w = 0, 0.1, 0.5, 1; 0.5 is an explicit reconstruction choice

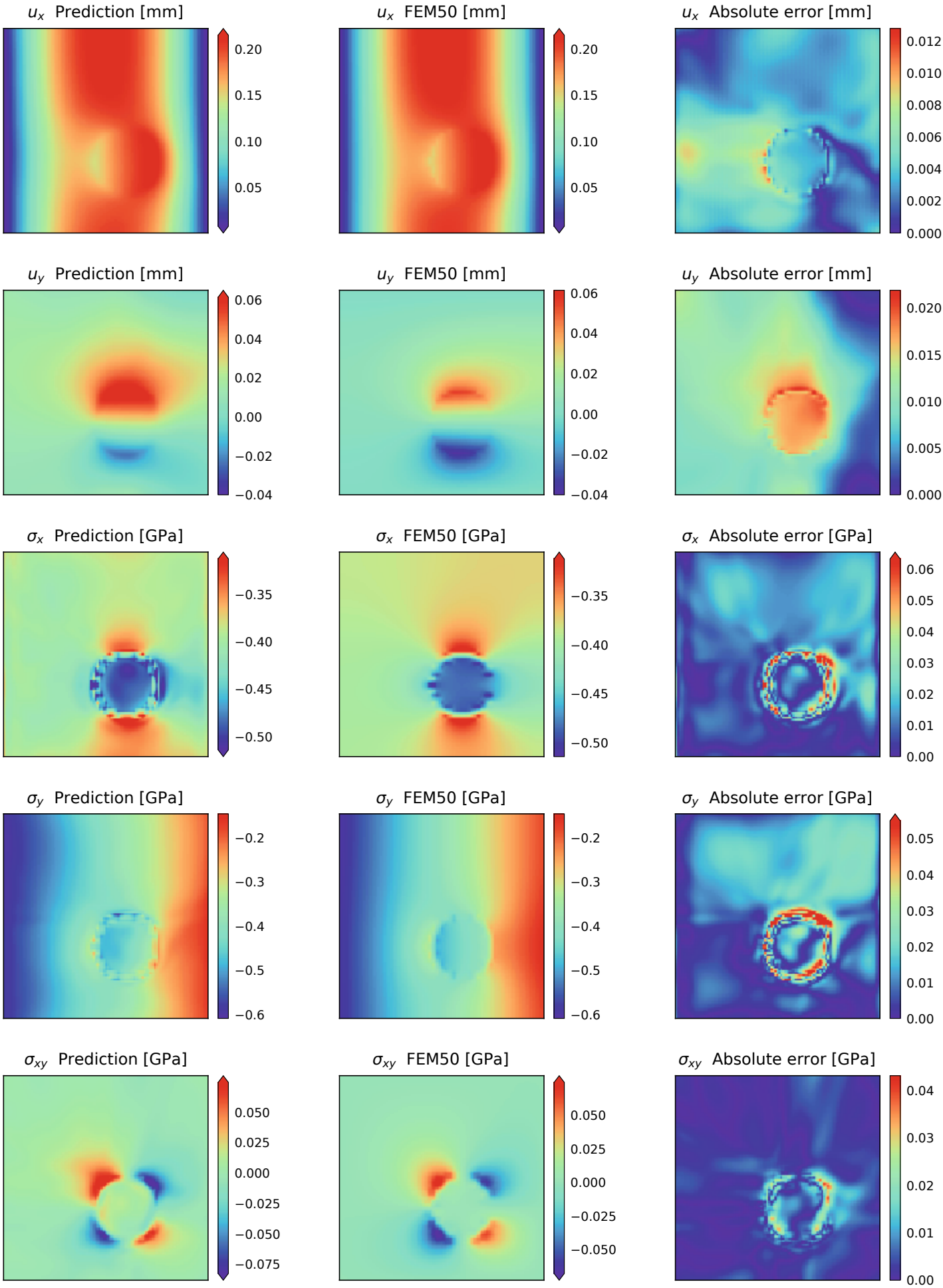


PARTIAL PROGRESS: completed jobs only | Train points reconstructed deterministically from saved config and sampler; evaluation coordinates read from prediction NPZ.

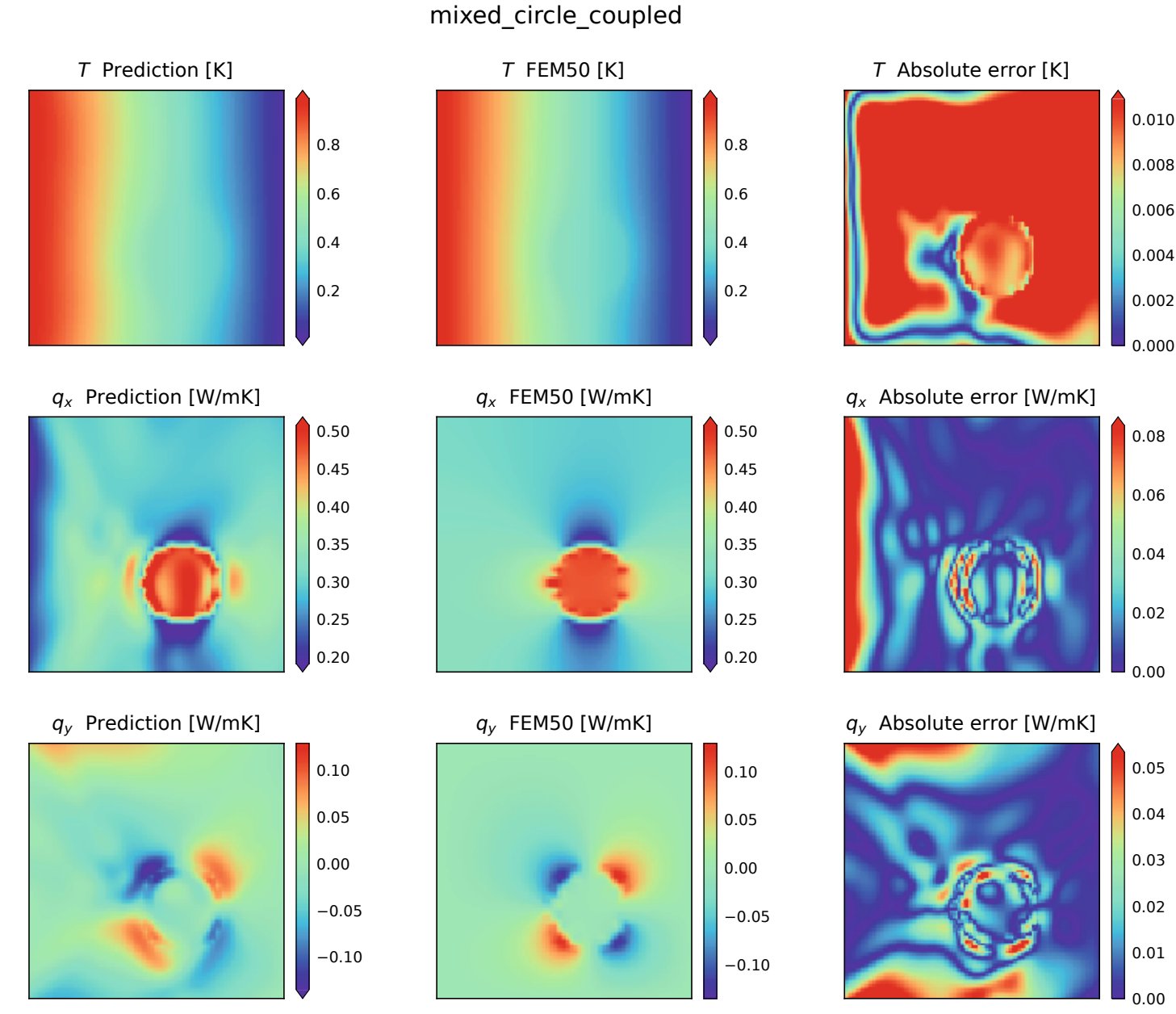


PARTIAL PROGRESS: completed jobs only | Displacement factor1; no fitted amplitude or coordinate clipping. Artificial benchmark alpha=1.

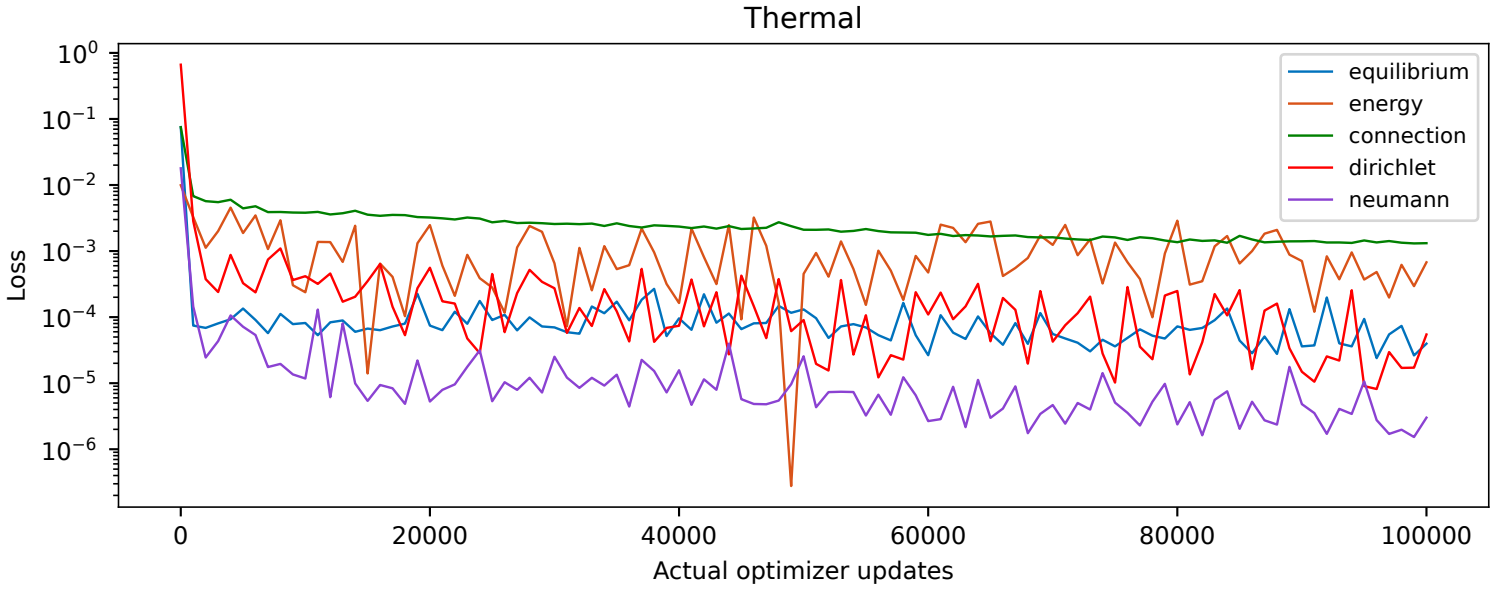
mixed_circle_coupled

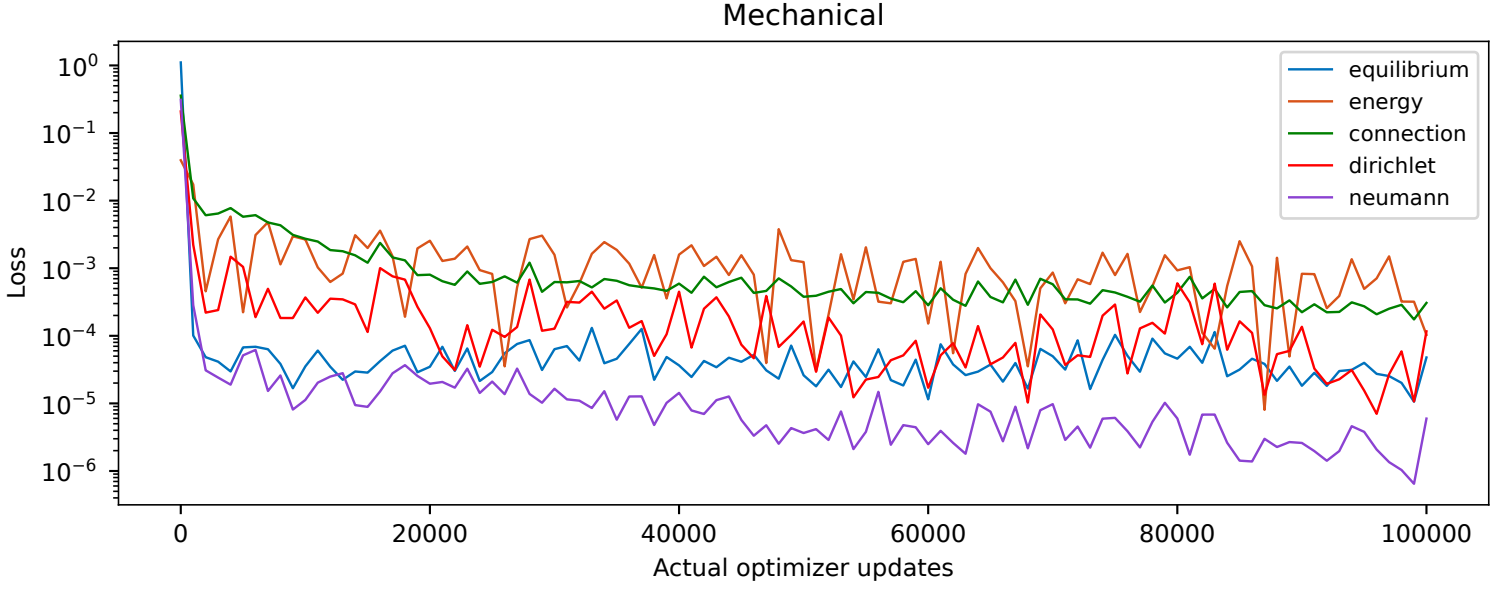


PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.



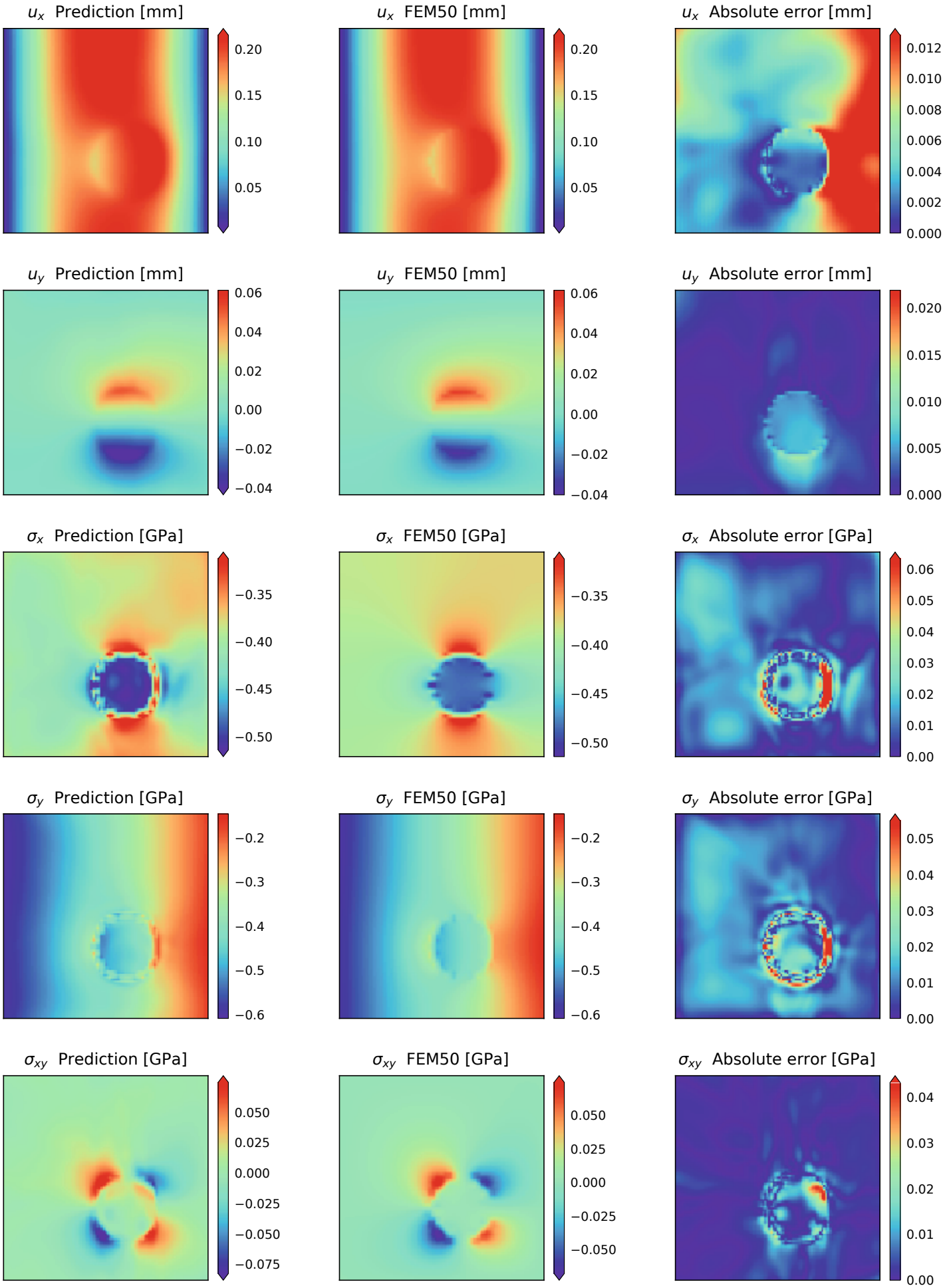
PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.



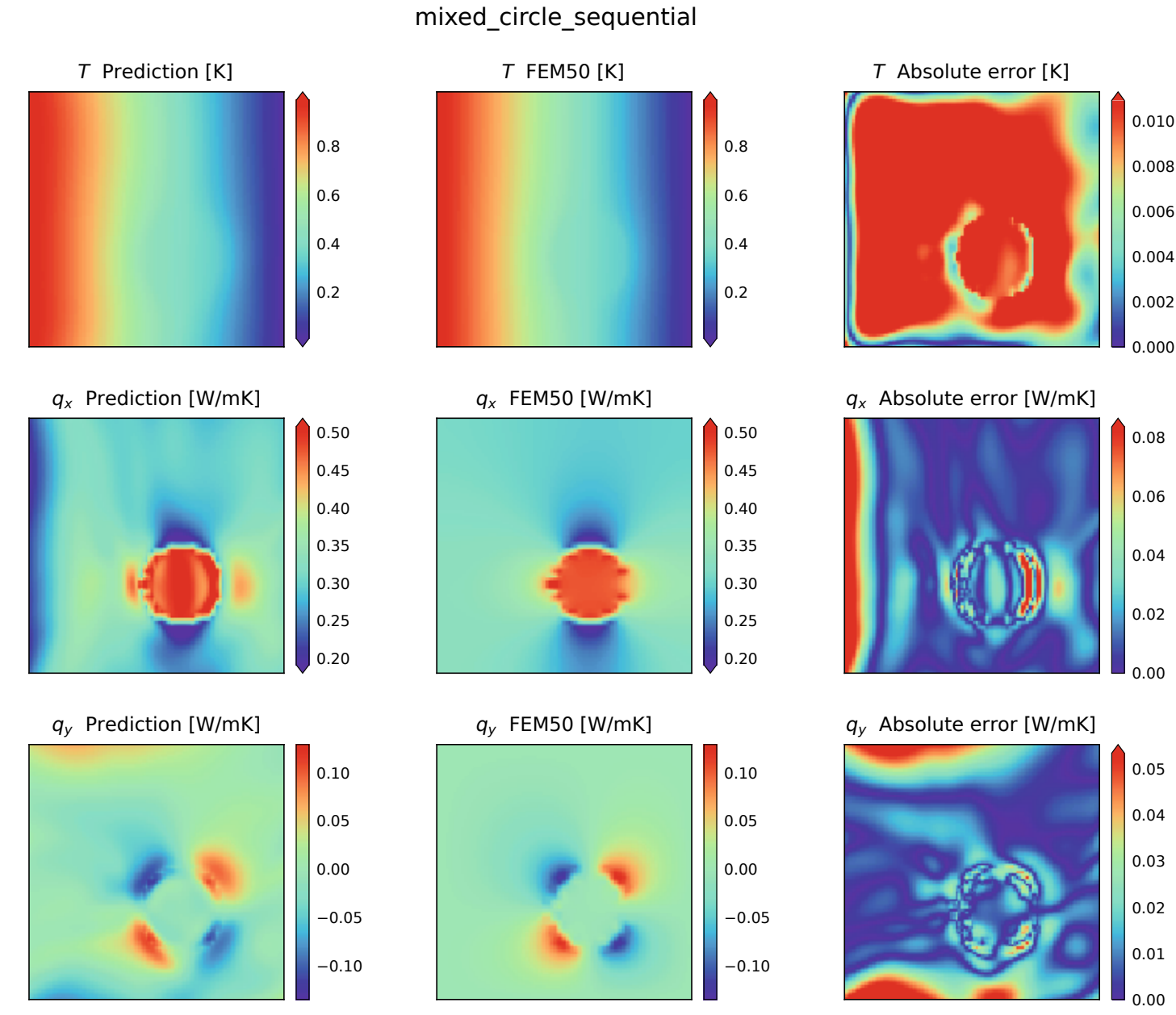


PARTIAL PROGRESS: completed jobs only | Original logged samples only. Zero terms omitted on log axes; inactive terms are not filled or bridged.

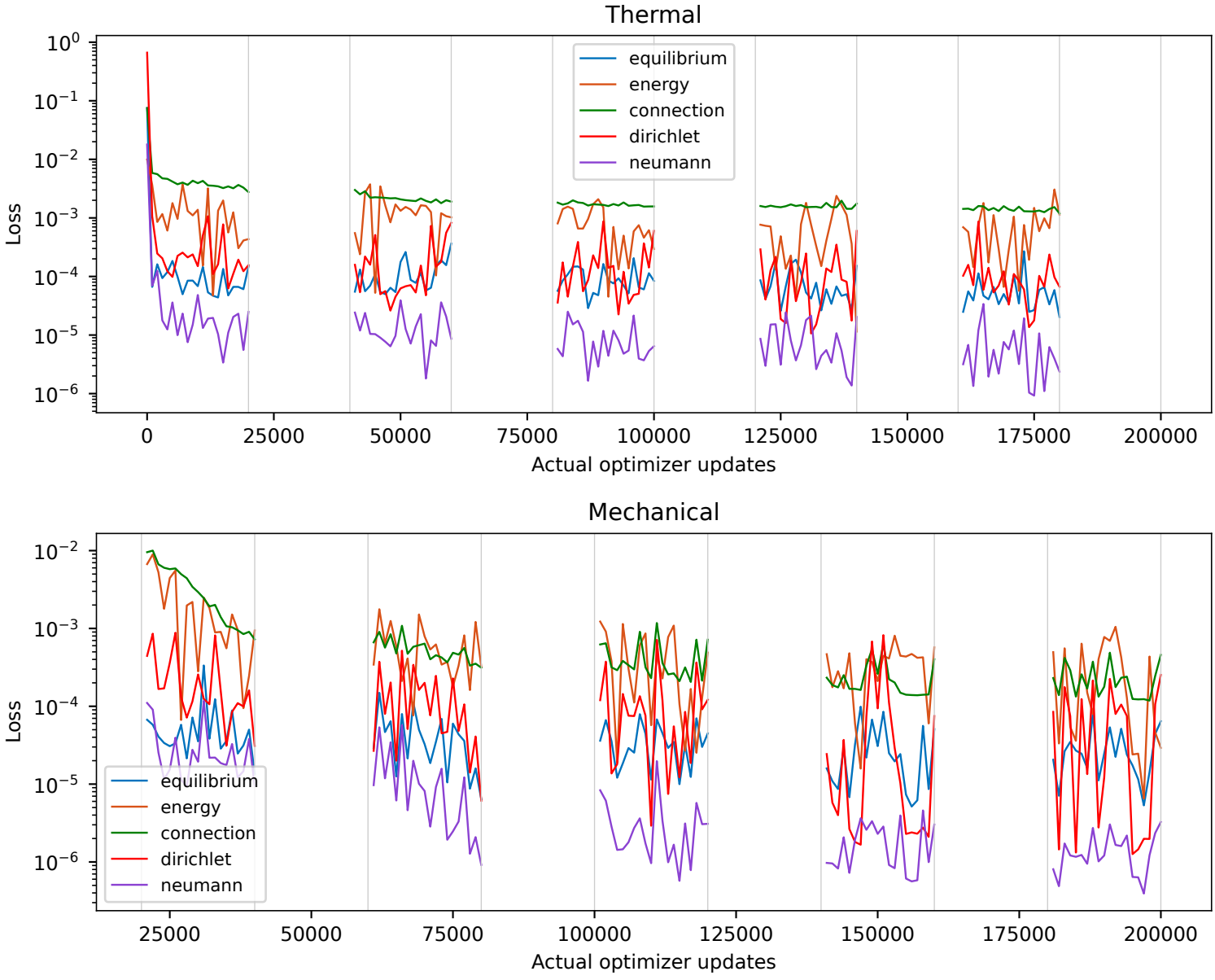
mixed_circle_sequential



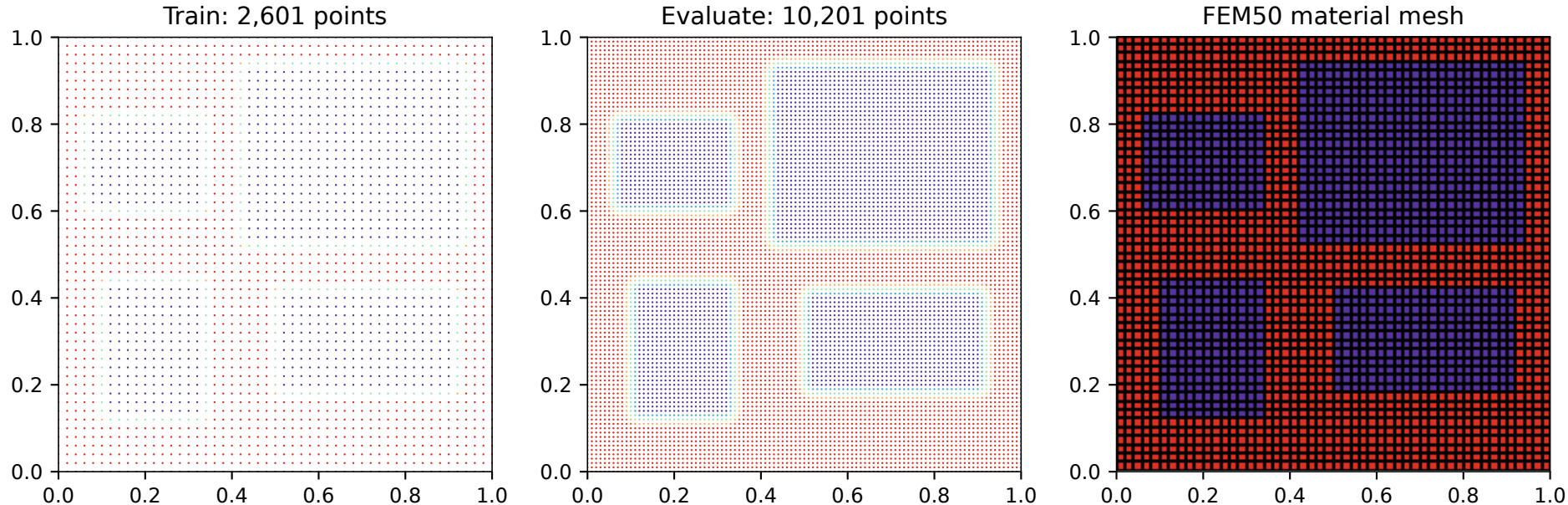
PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.



PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

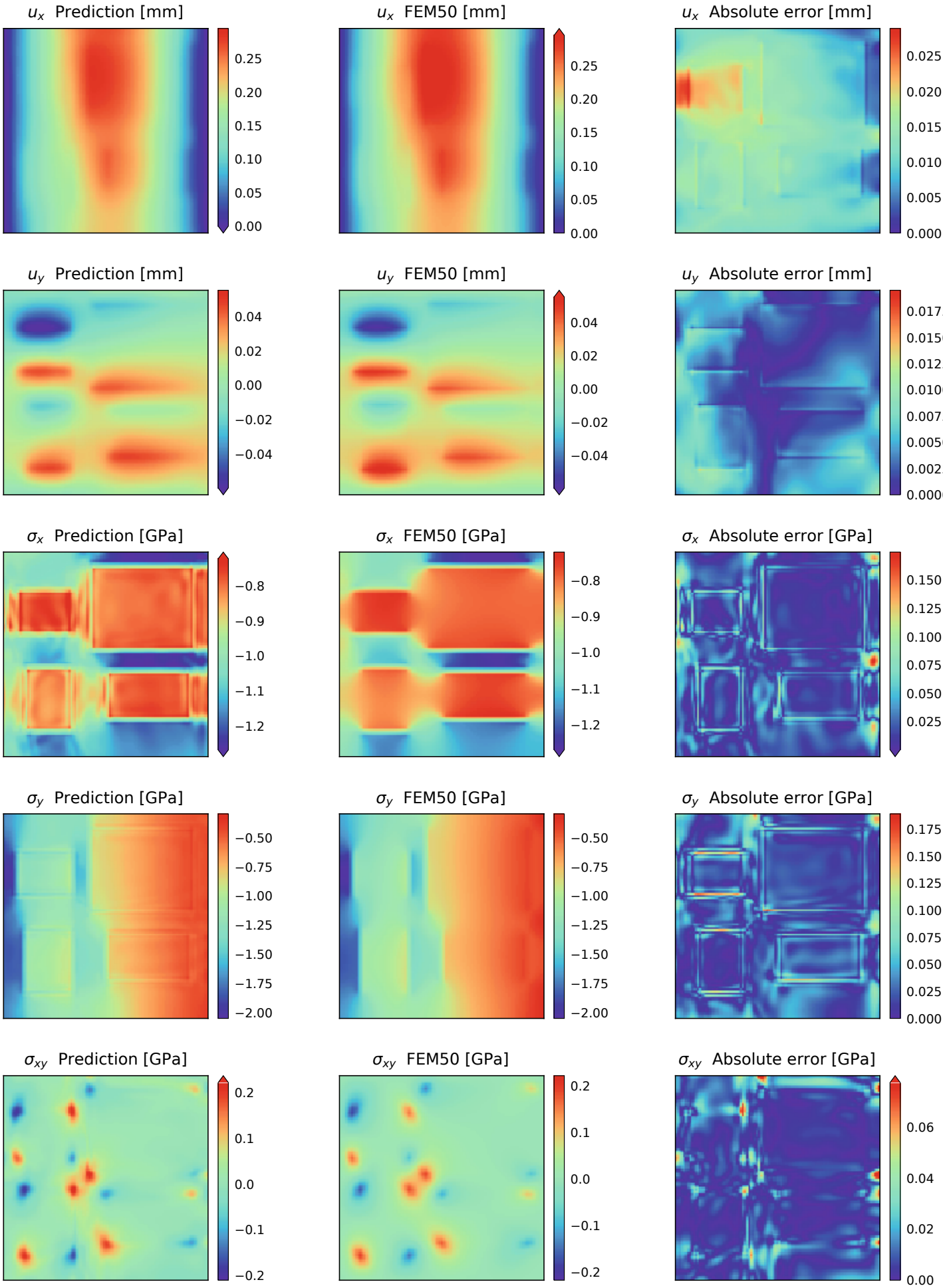


PARTIAL PROGRESS: completed jobs only | Original logged samples only. Zero terms omitted on log axes; inactive terms are not filled or bridged.

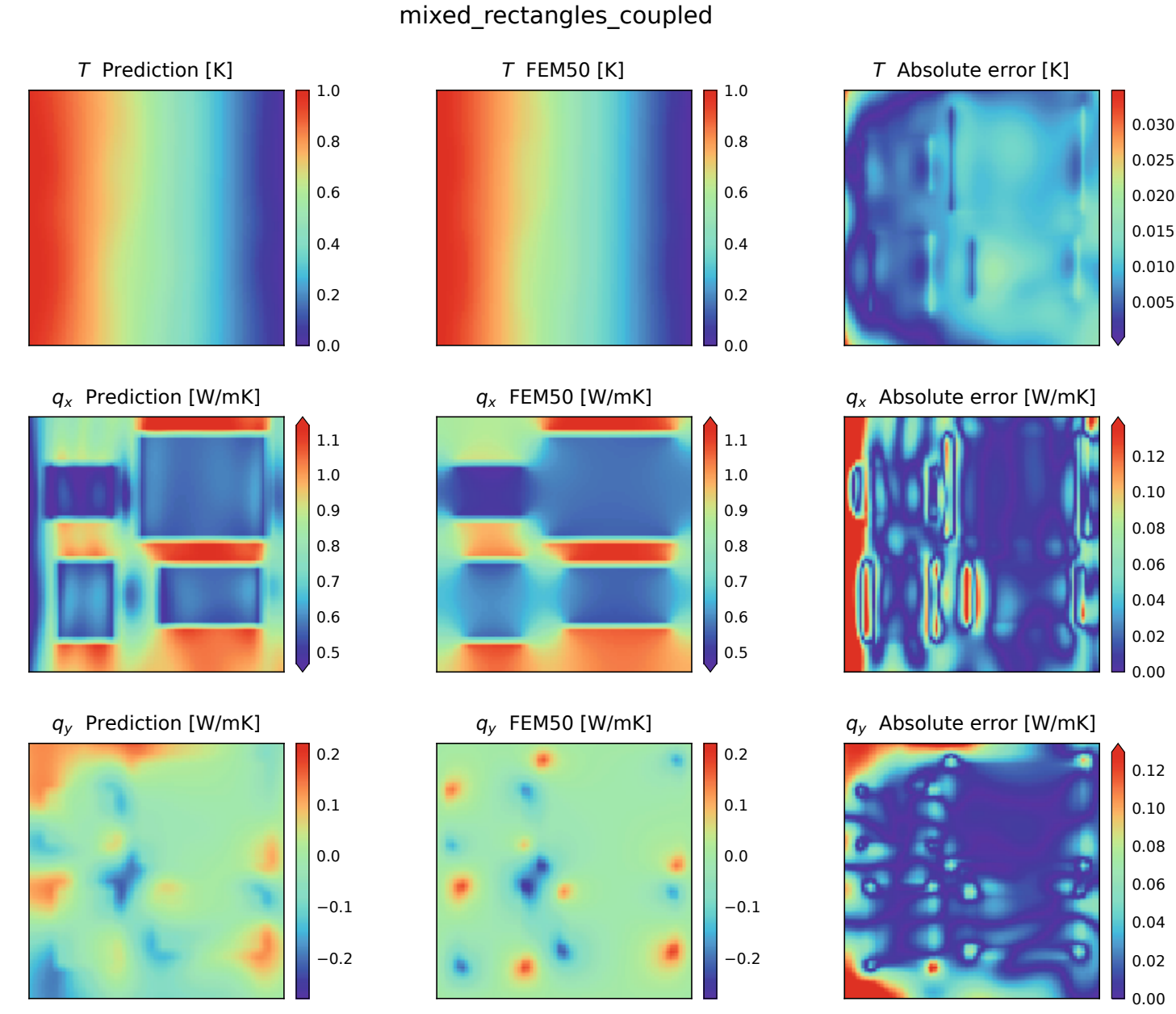


PARTIAL PROGRESS: completed jobs only | Train points reconstructed deterministically from saved config and sampler; evaluation coordinates read from prediction NPZ.

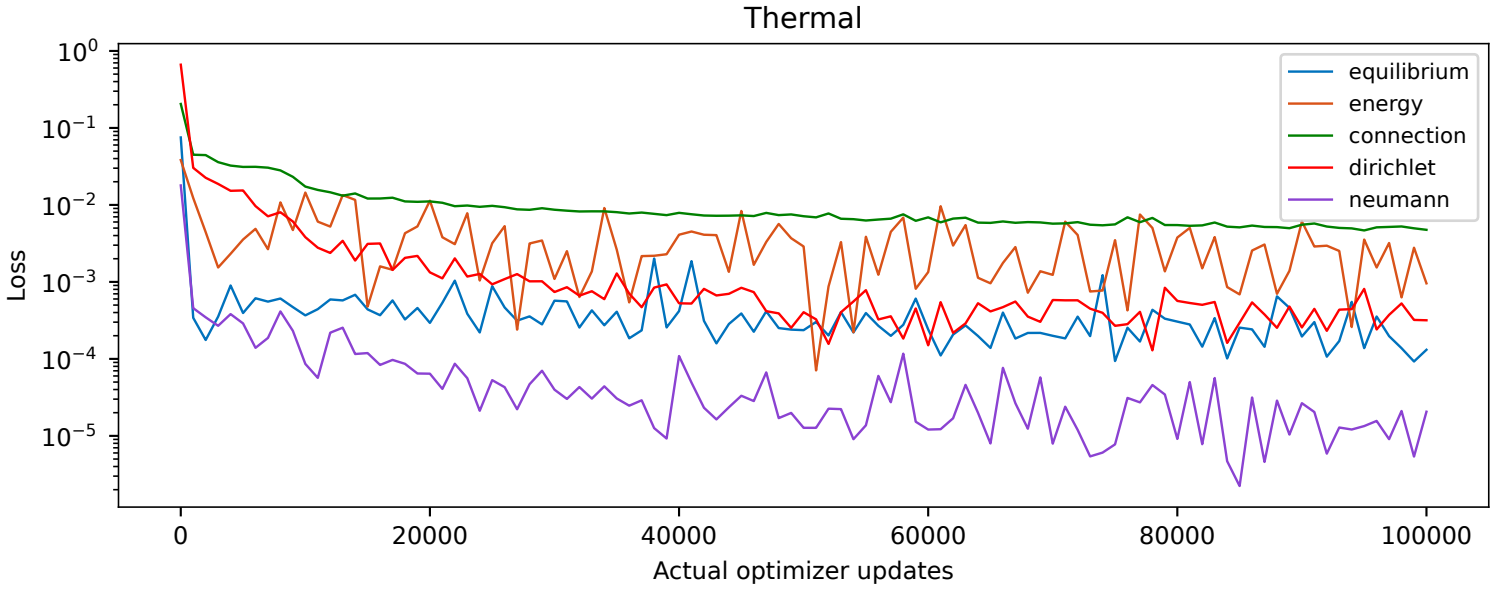
mixed_rectangles_coupled

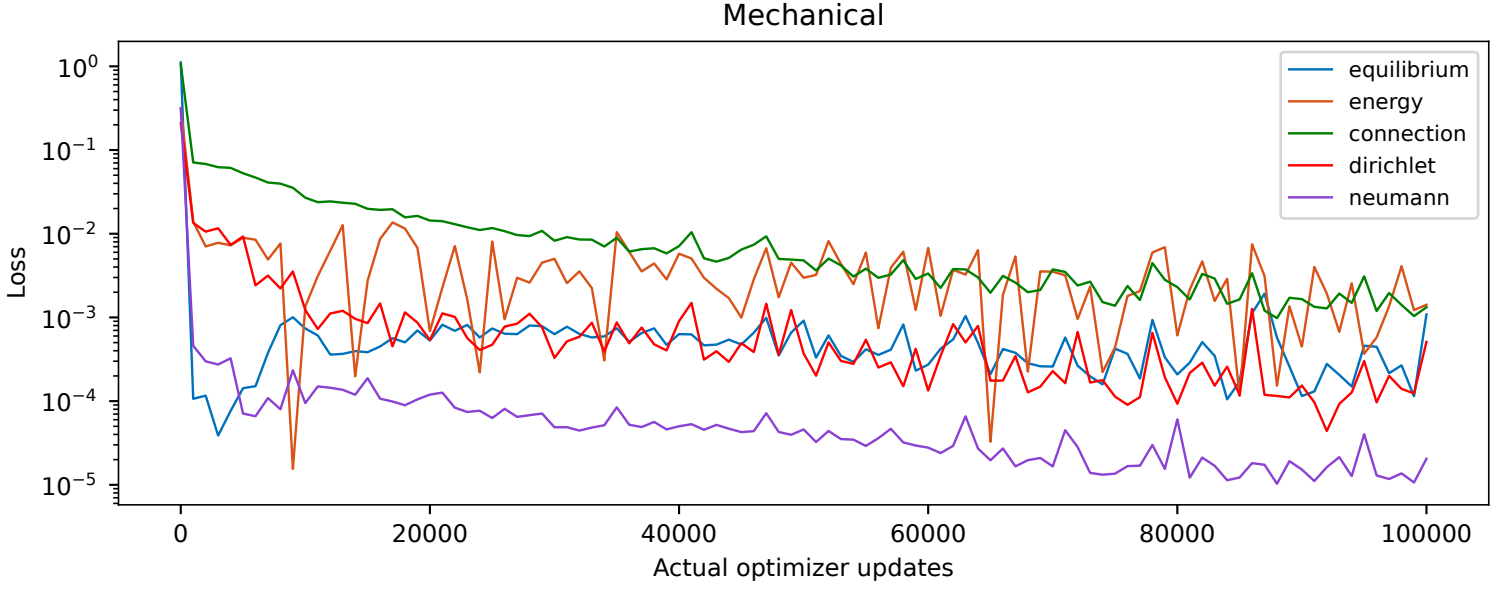


PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.



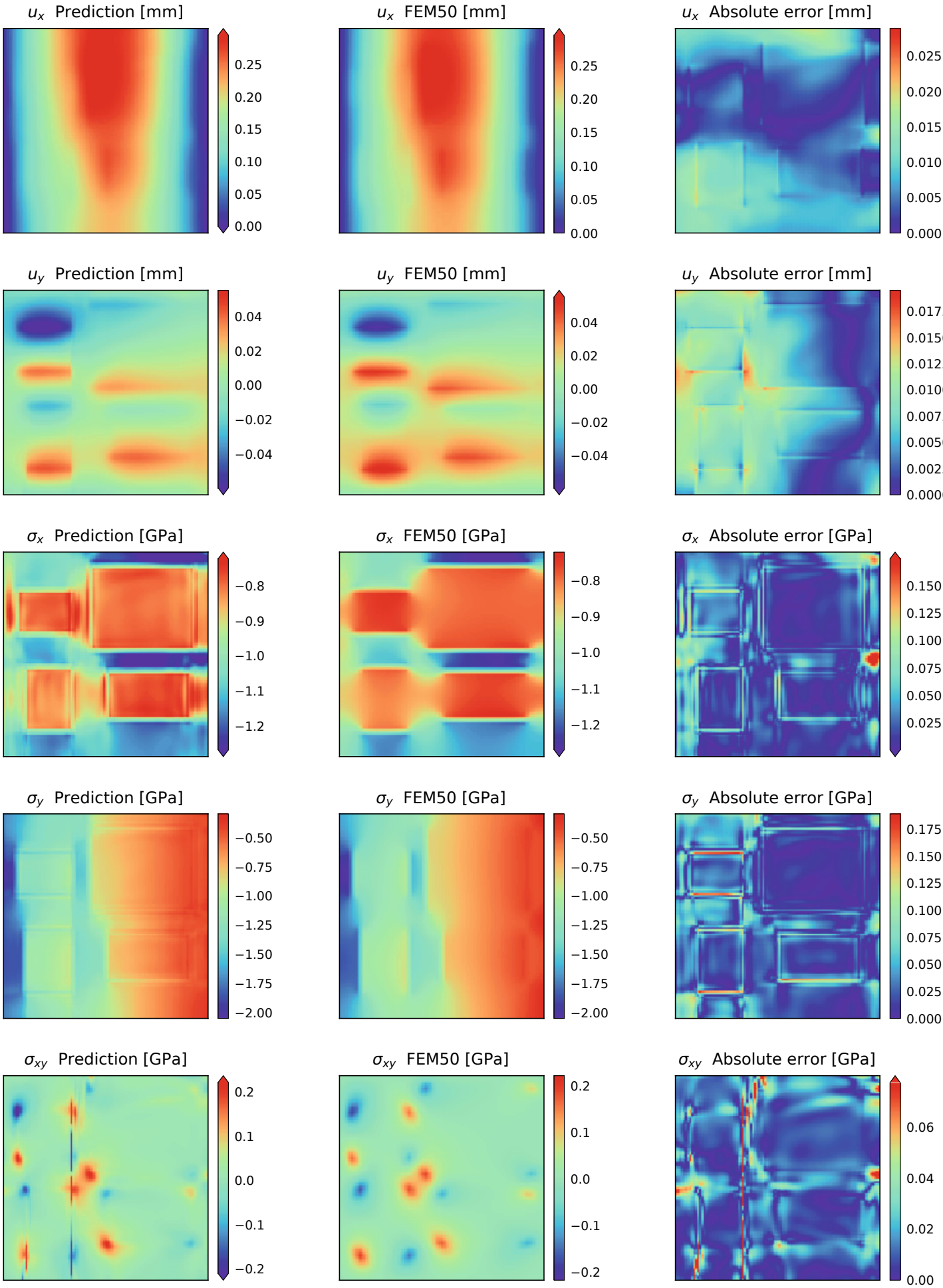
PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.



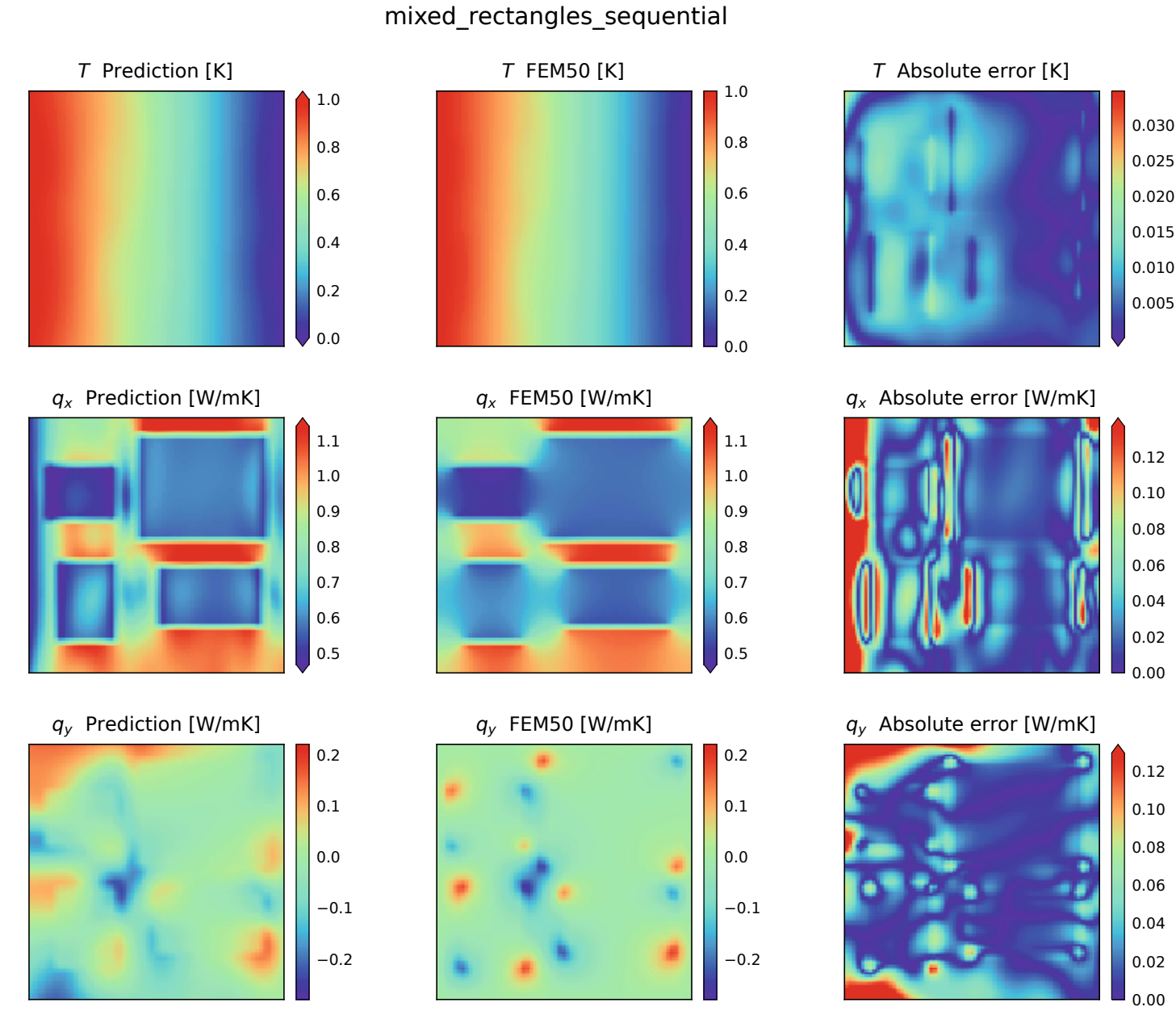


PARTIAL PROGRESS: completed jobs only | Original logged samples only. Zero terms omitted on log axes; inactive terms are not filled or bridged.

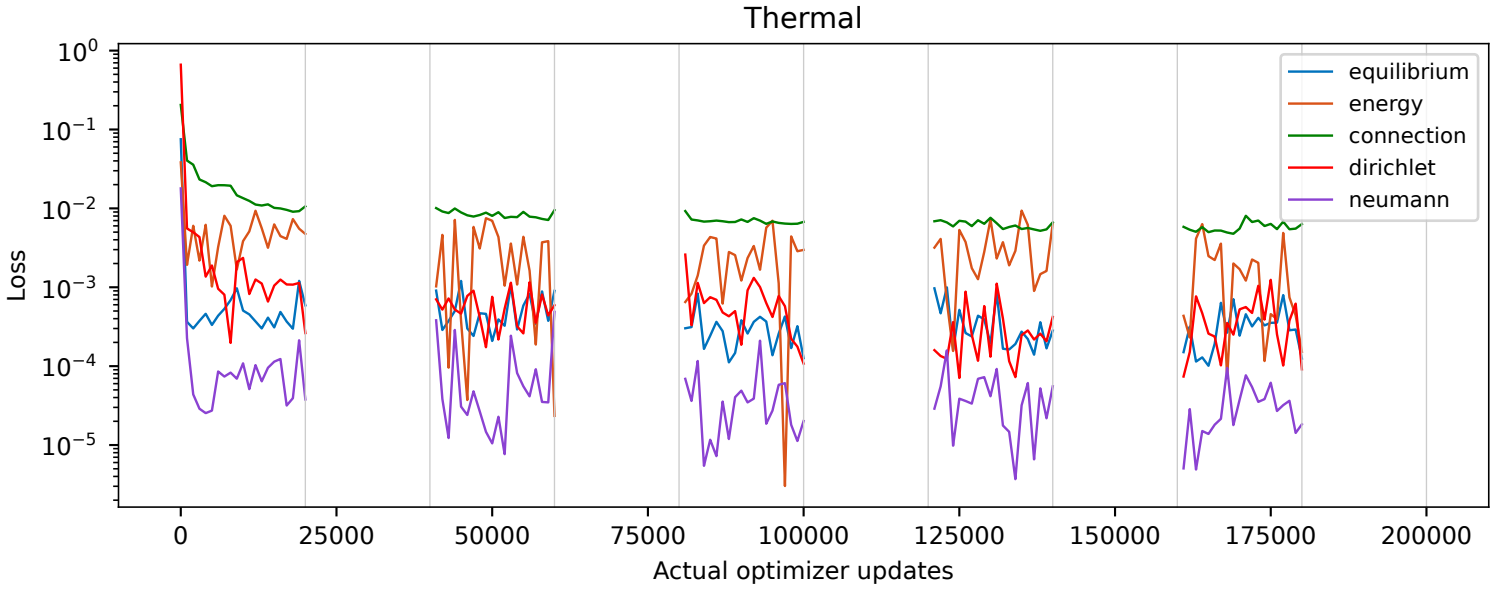
mixed_rectangles_sequential

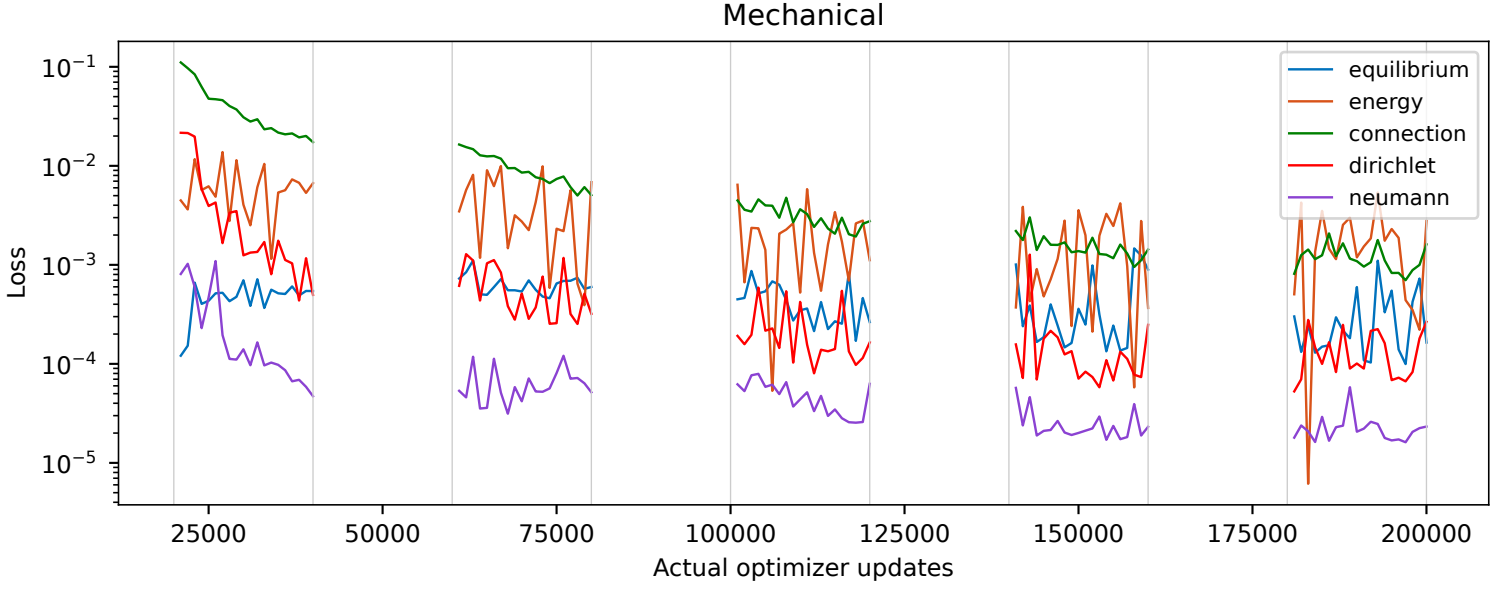


PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

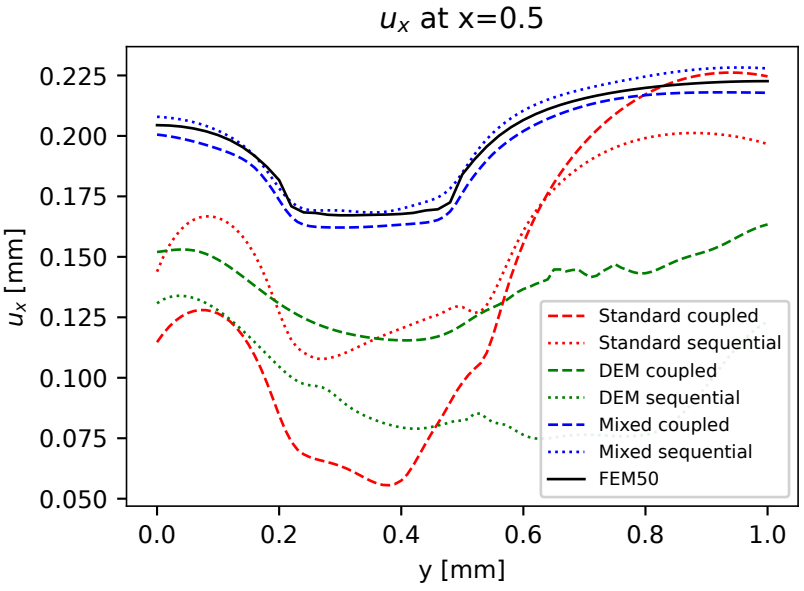


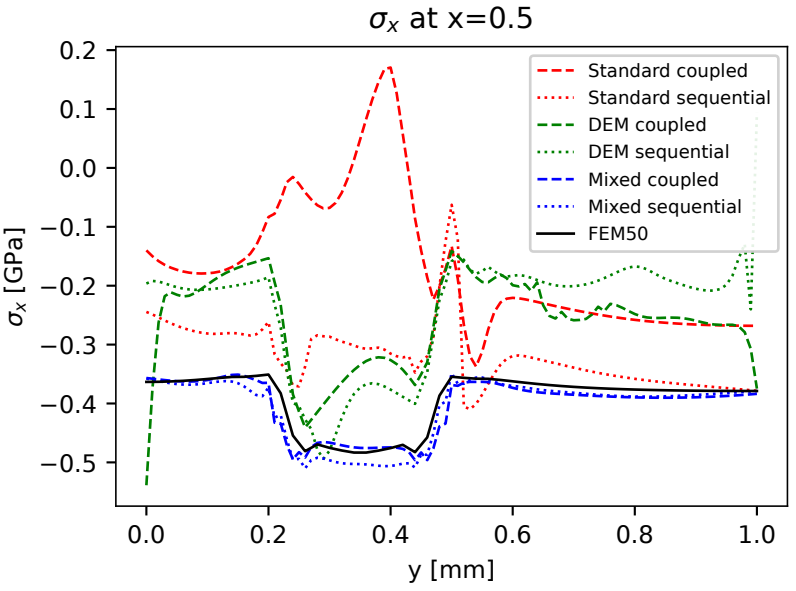
PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

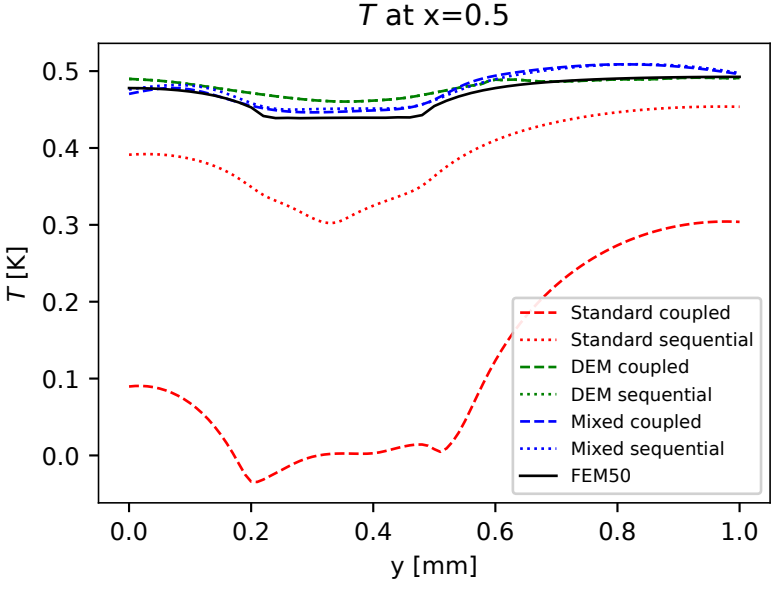


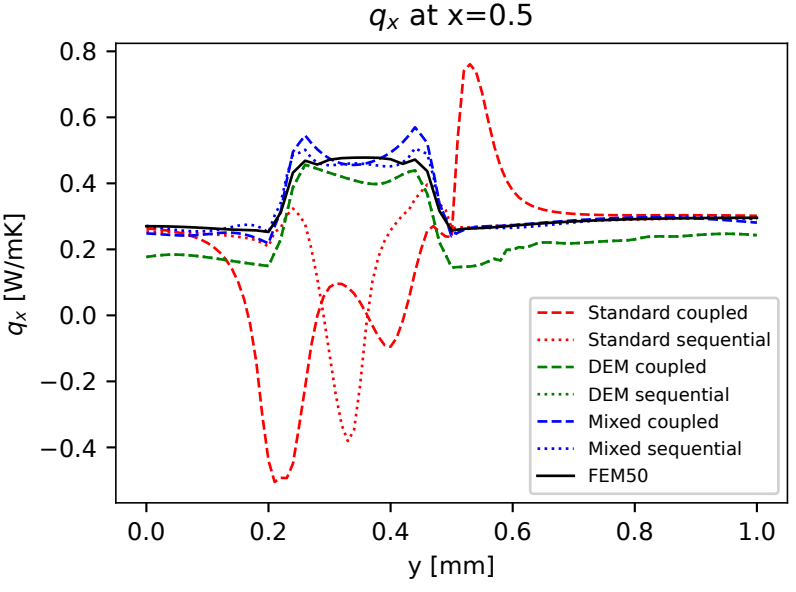


PARTIAL PROGRESS: completed jobs only | Original logged samples only. Zero terms omitted on log axes; inactive terms are not filled or bridged.

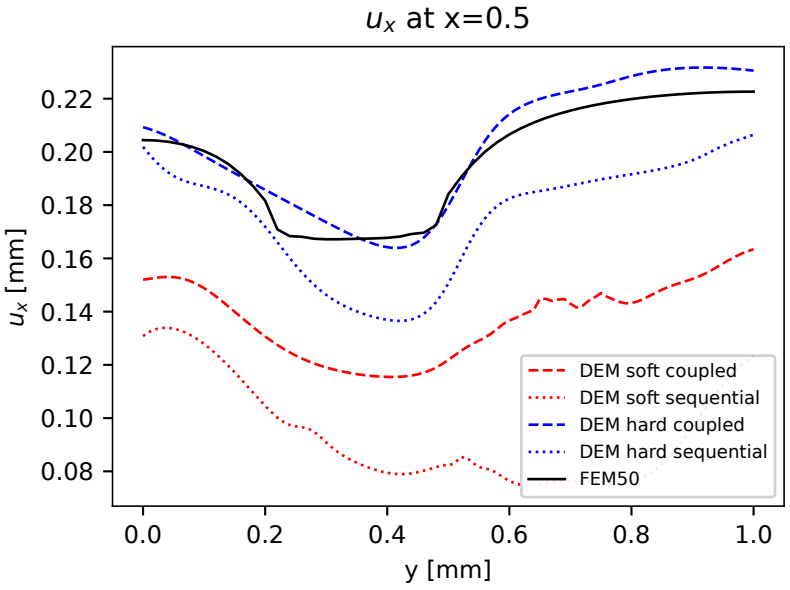


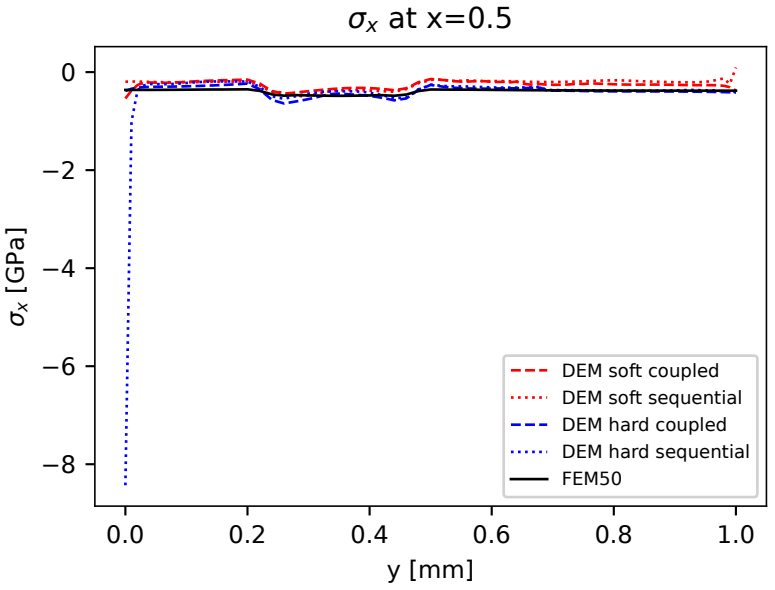


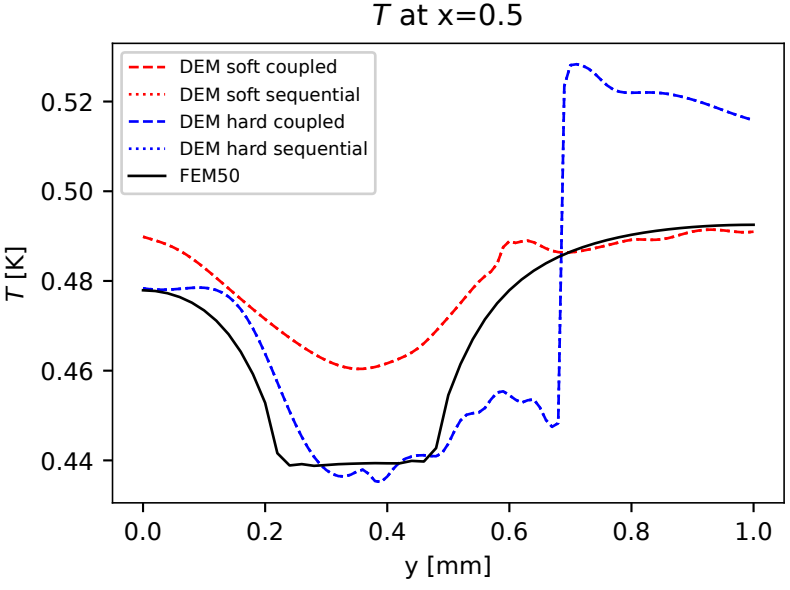


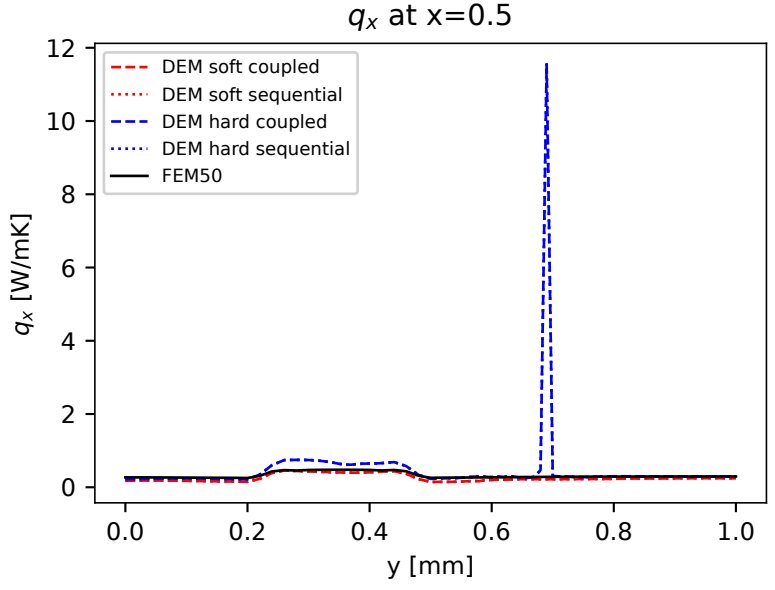


PARTIAL PROGRESS: completed jobs only | All six completed full-budget cases, x=.5. Negative outcomes shown without ranking assumptions.

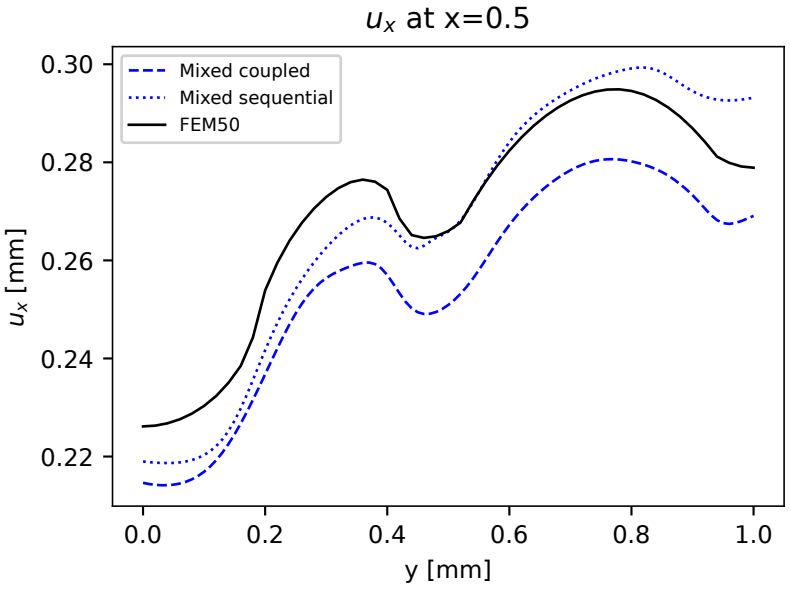


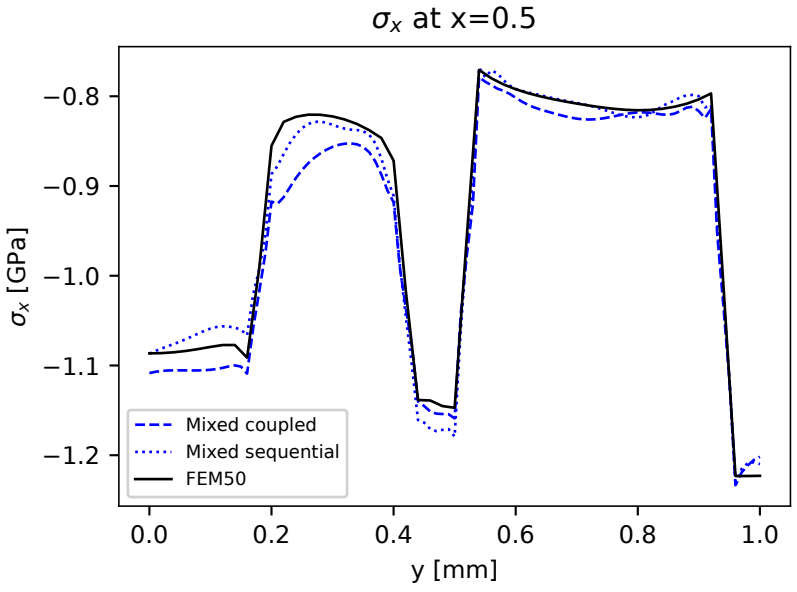


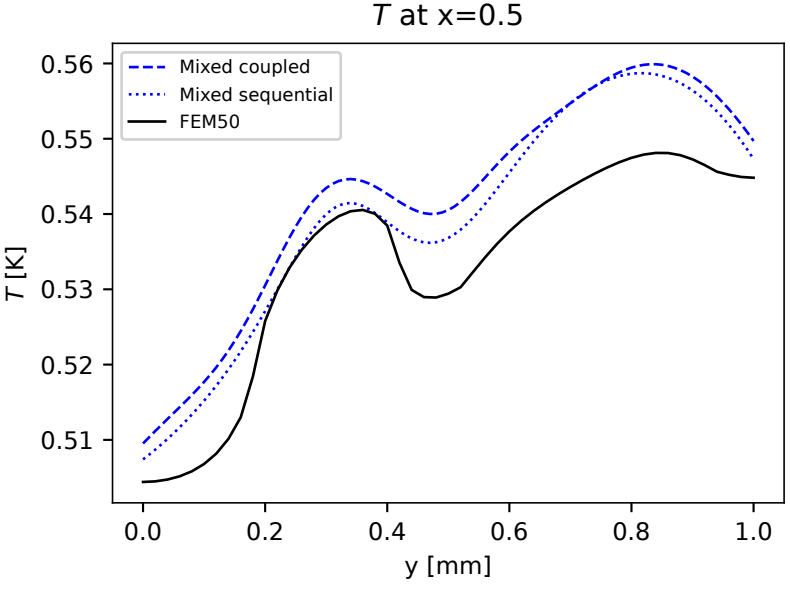


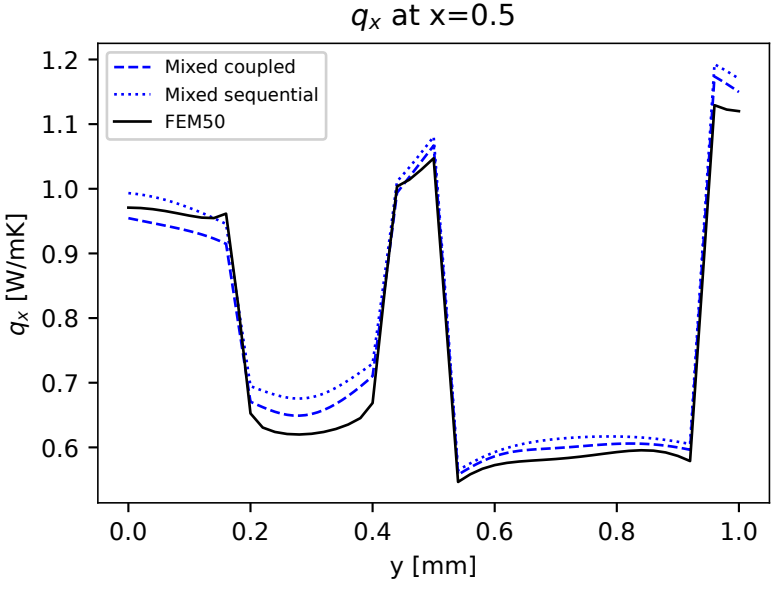


PARTIAL PROGRESS: completed jobs only | Additional DEM soft/hard boundary sensitivity; not a separately reported manuscript experiment.

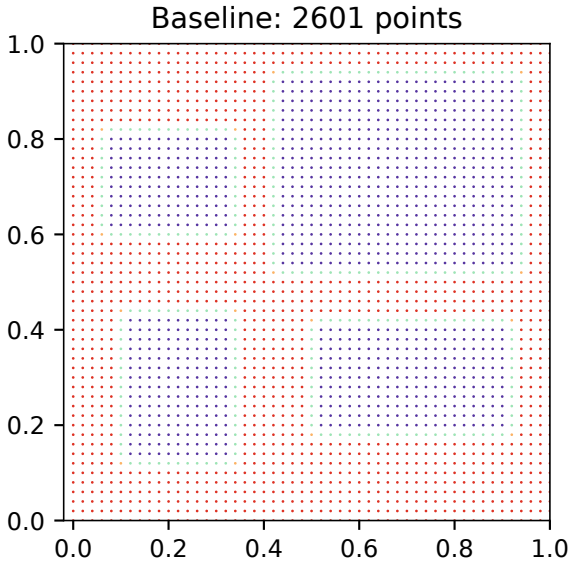


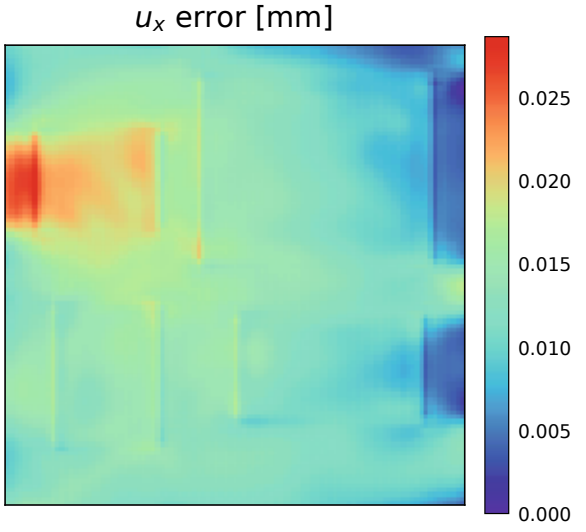


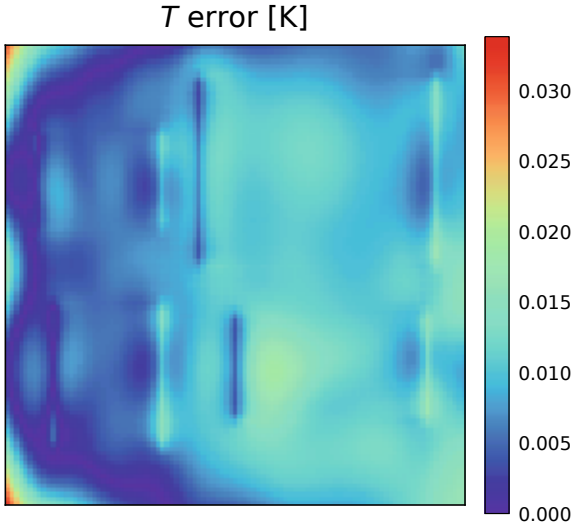


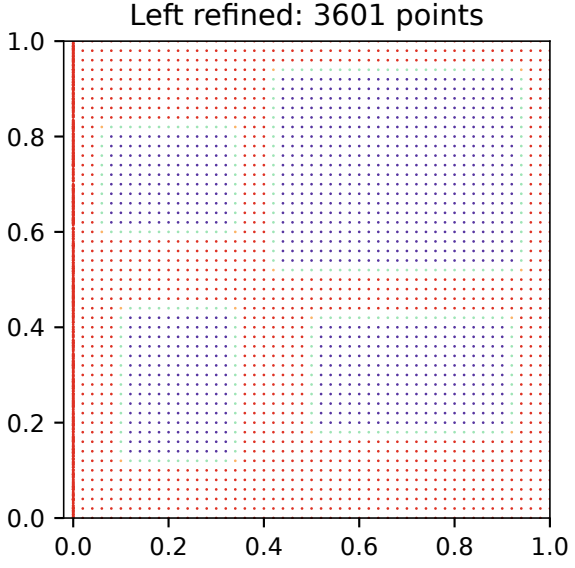


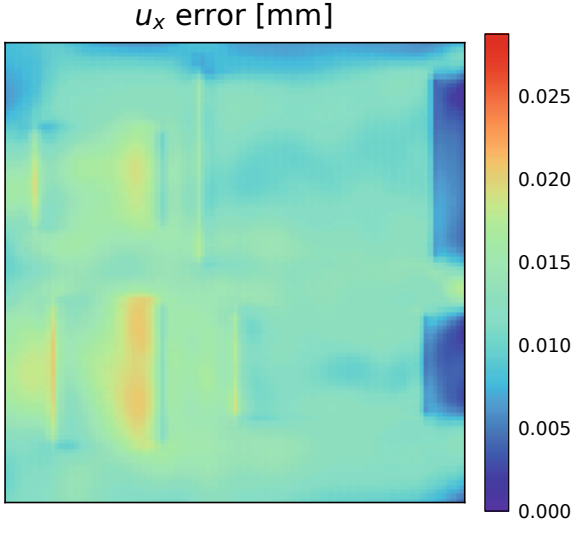
PARTIAL PROGRESS: completed jobs only | Saved full-budget coupled and sequential predictions; no fitted scaling.

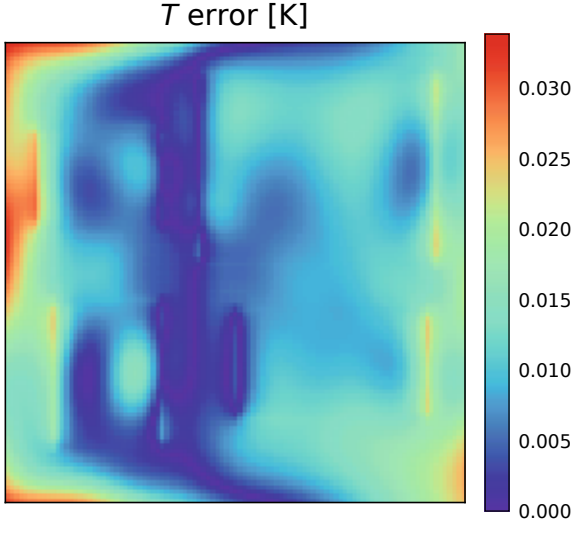




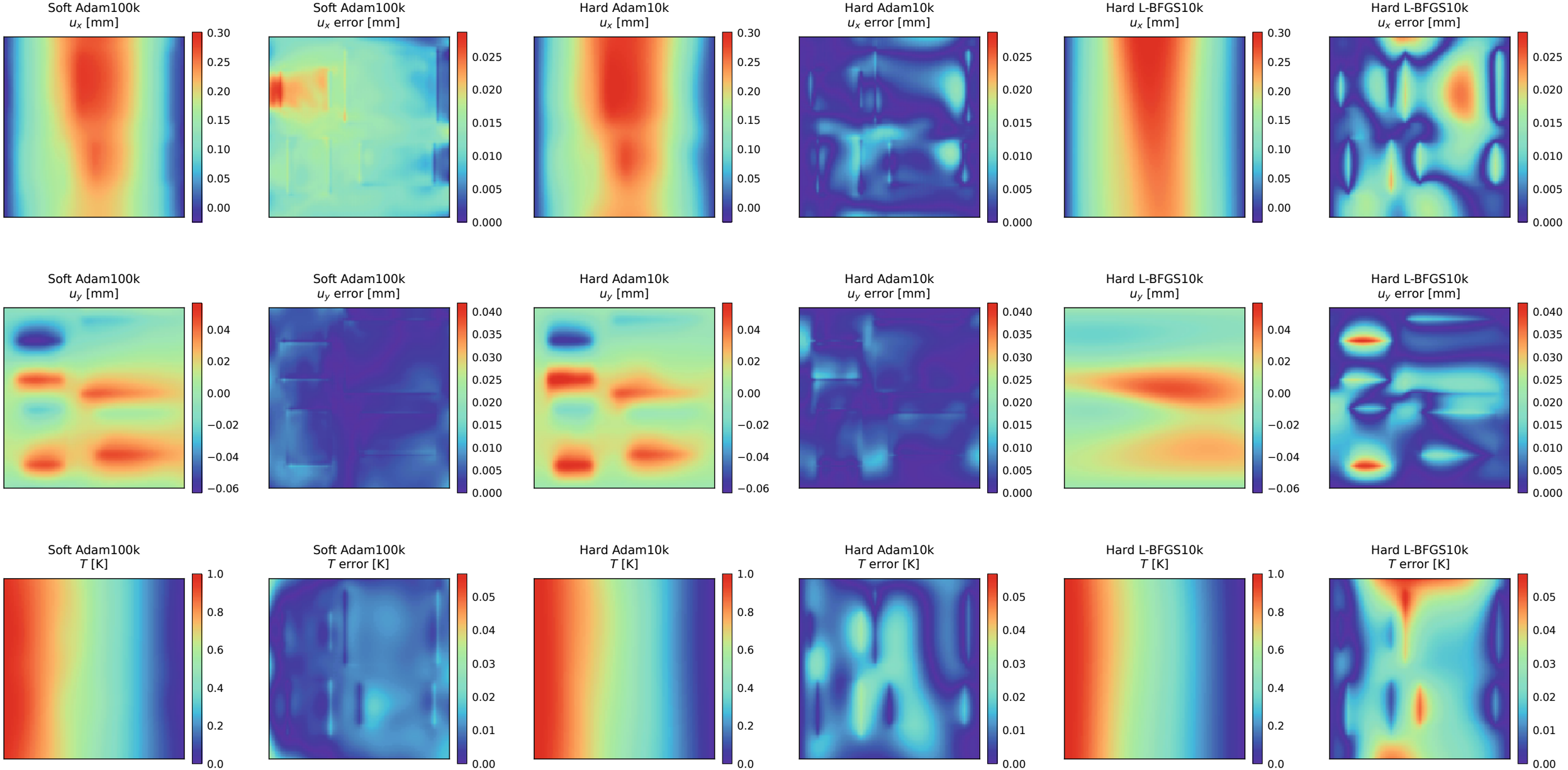




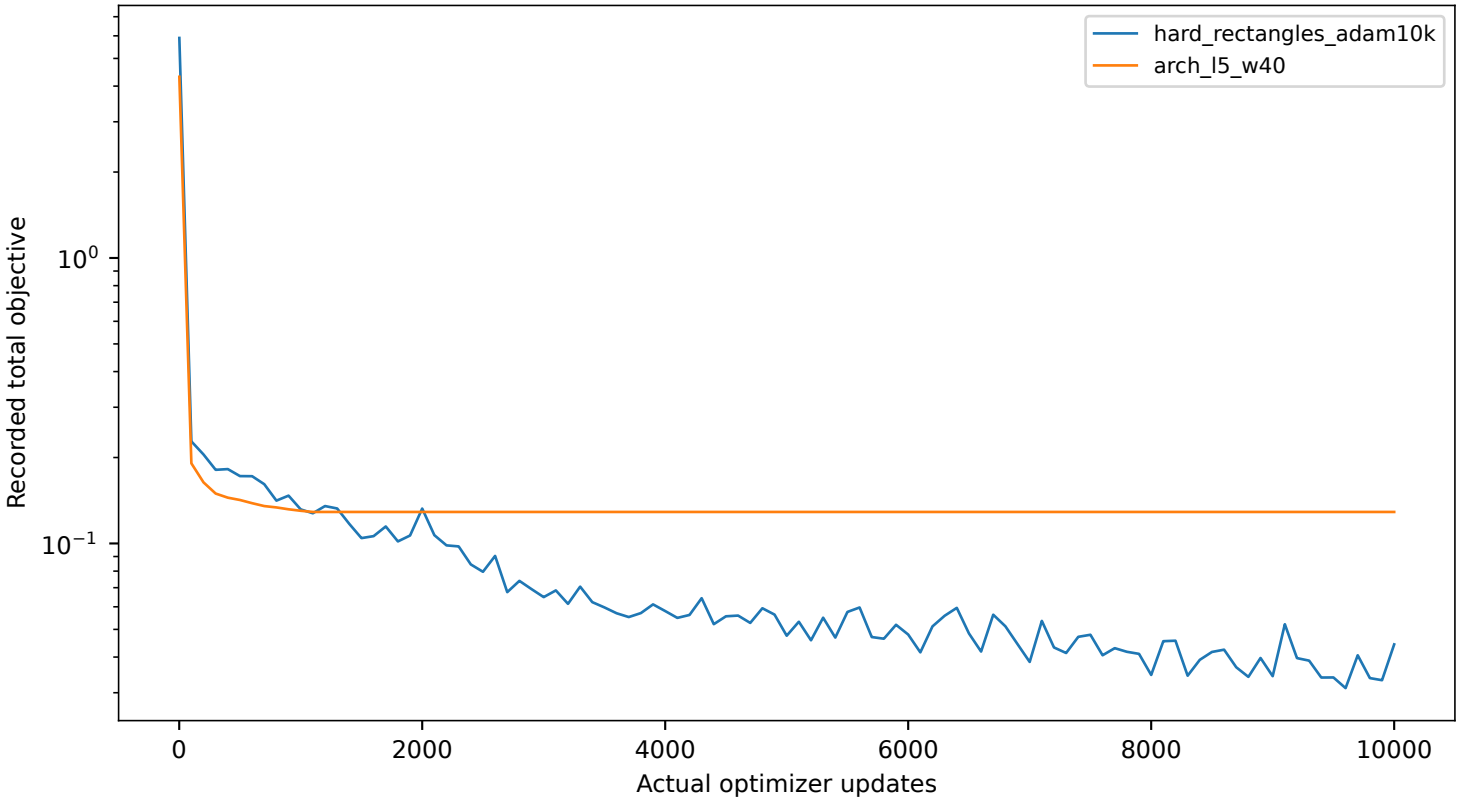




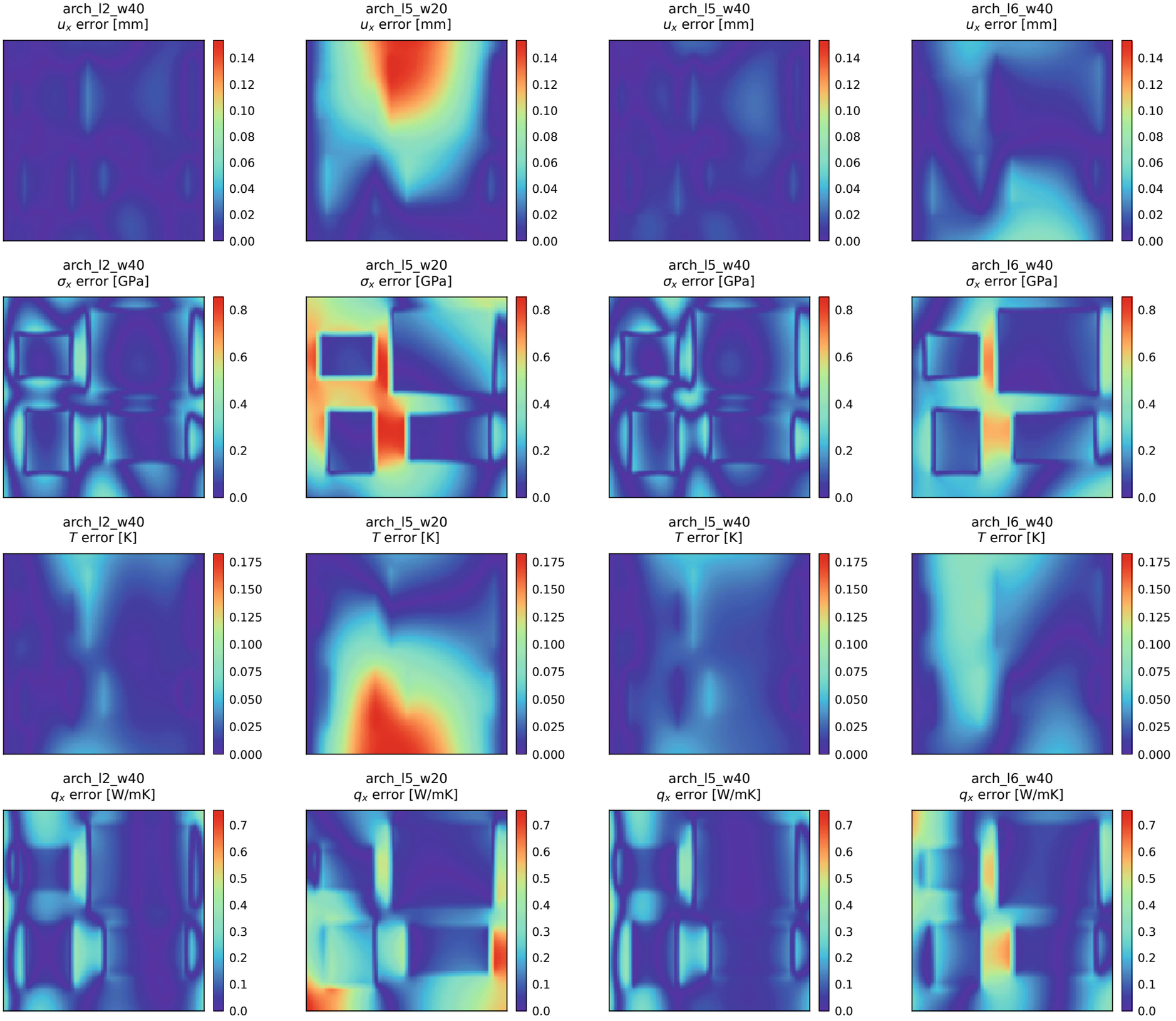
PARTIAL PROGRESS: completed jobs only | Same deterministic sampler;1000left-edge points added. Shared error scales; actual outcomes may improve or deteriorate.



PARTIAL PROGRESS: completed jobs only | Field/error pairs. Soft Adam100k versus hard Adam10k and hard L-BFGS10k; unequal budgets disclosed.

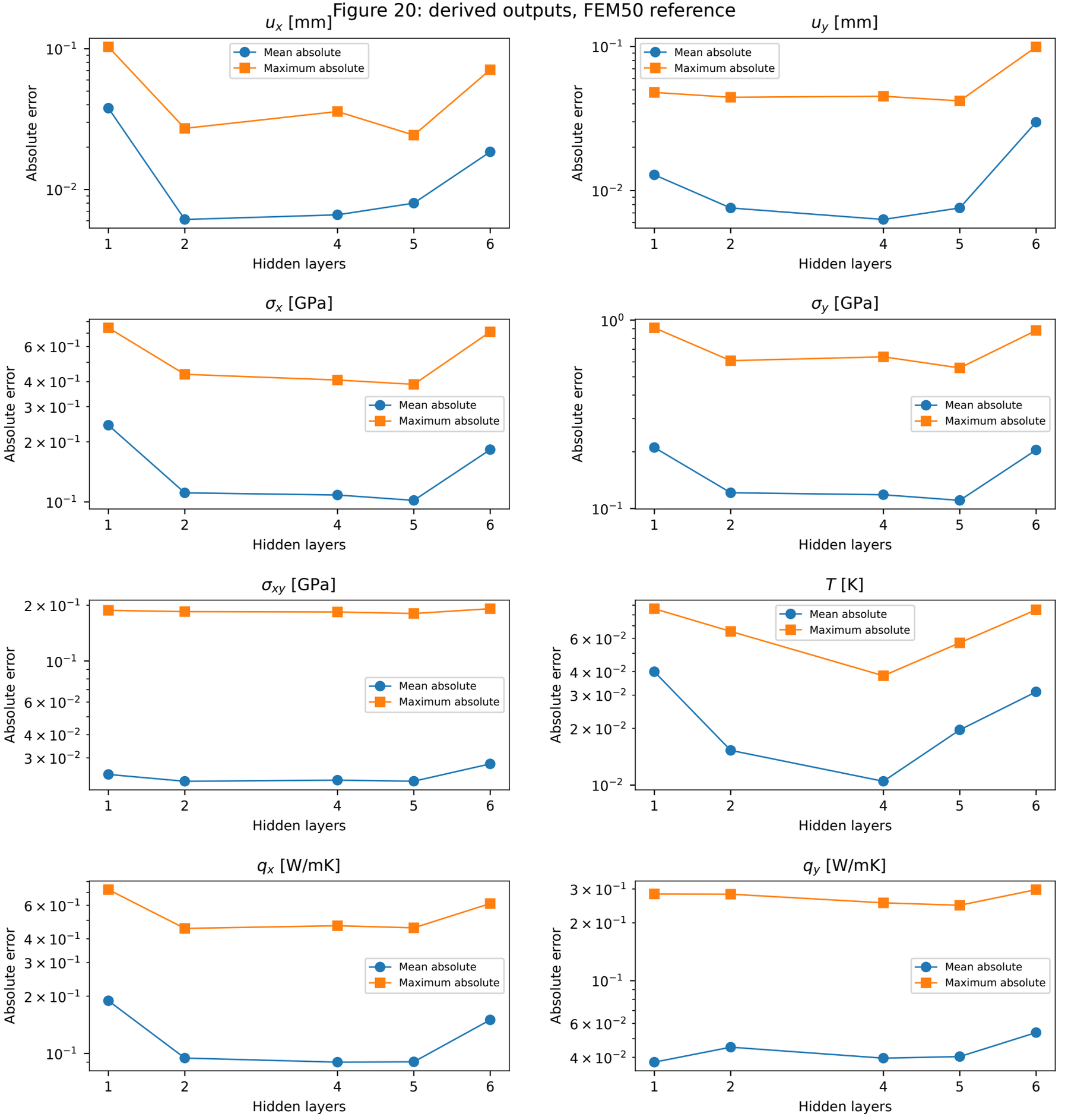


PARTIAL PROGRESS: completed jobs only | Original logged objectives; Adam and L-BFGS logging evaluation moments may differ. No resampling.



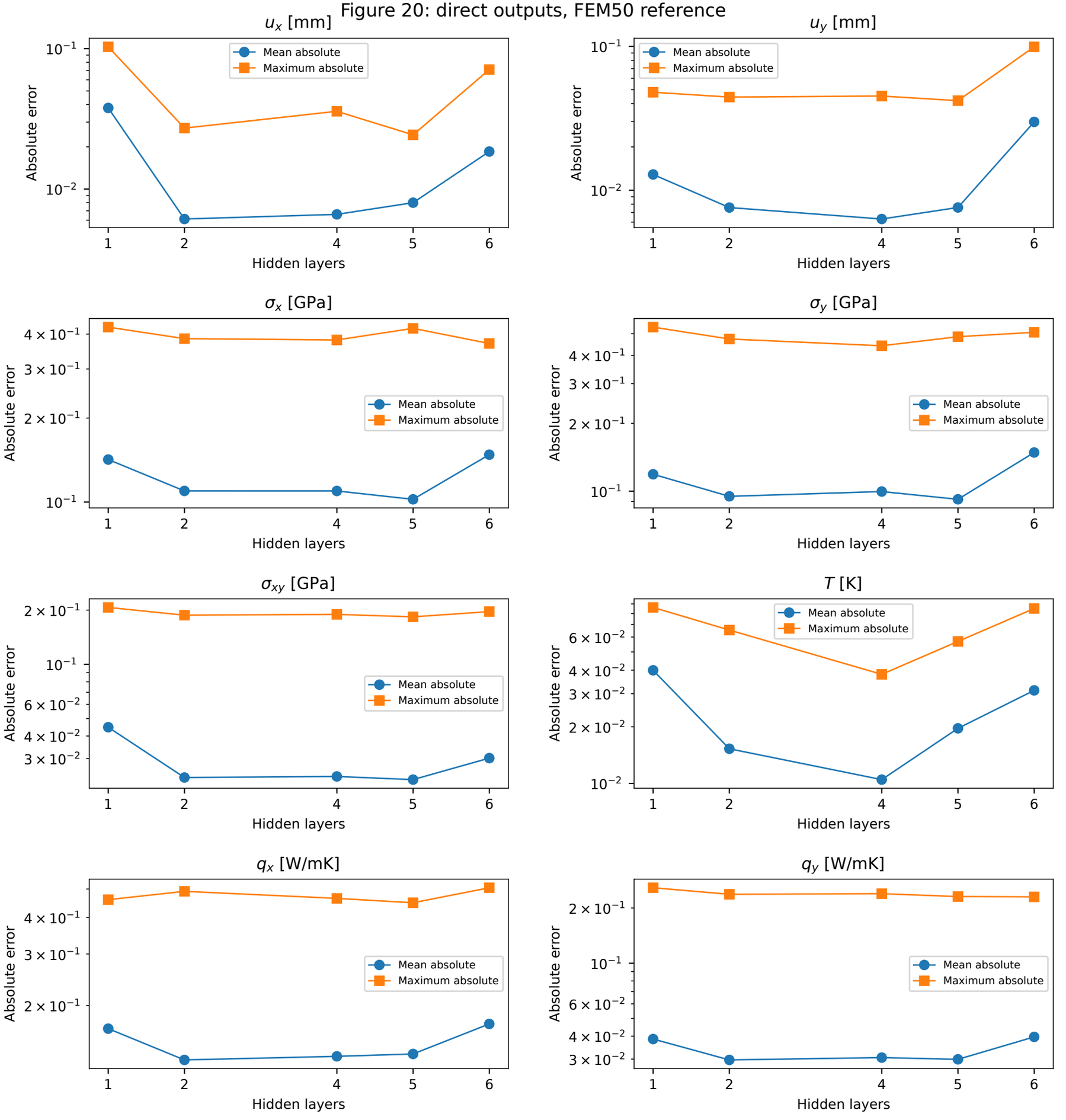
PARTIAL PROGRESS: completed jobs only | Selected2x40,5x20,5x40,6x40; each10k L-BFGS. Shared per-field limits across cases; derived fields.

Figure 20: derived outputs, FEM50 reference



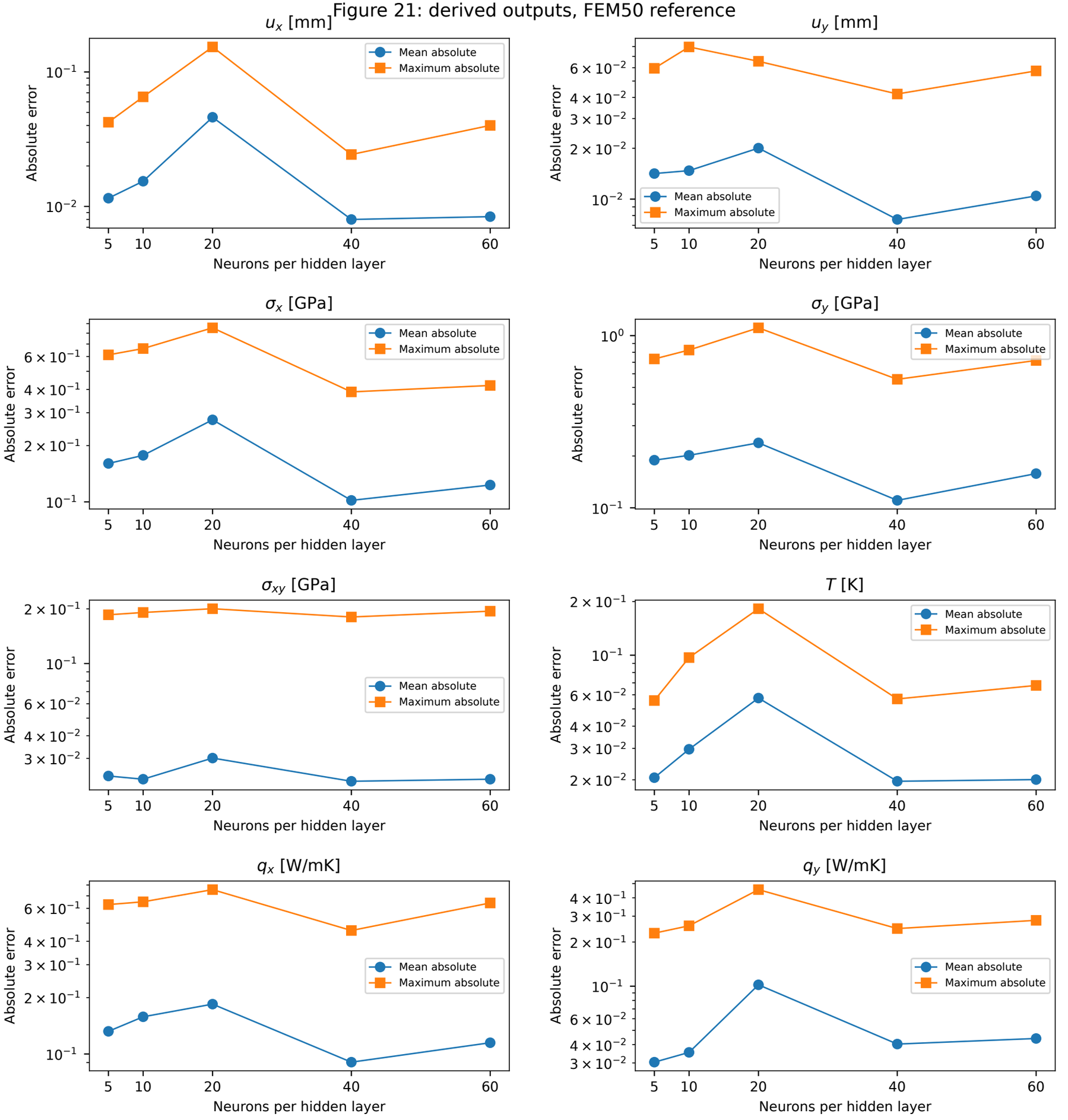
PARTIAL PROGRESS: completed jobs only | All eight fields. Maximum and mean measured errors; no assumption of monotonic improvement.

Figure 20: direct outputs, FEM50 reference



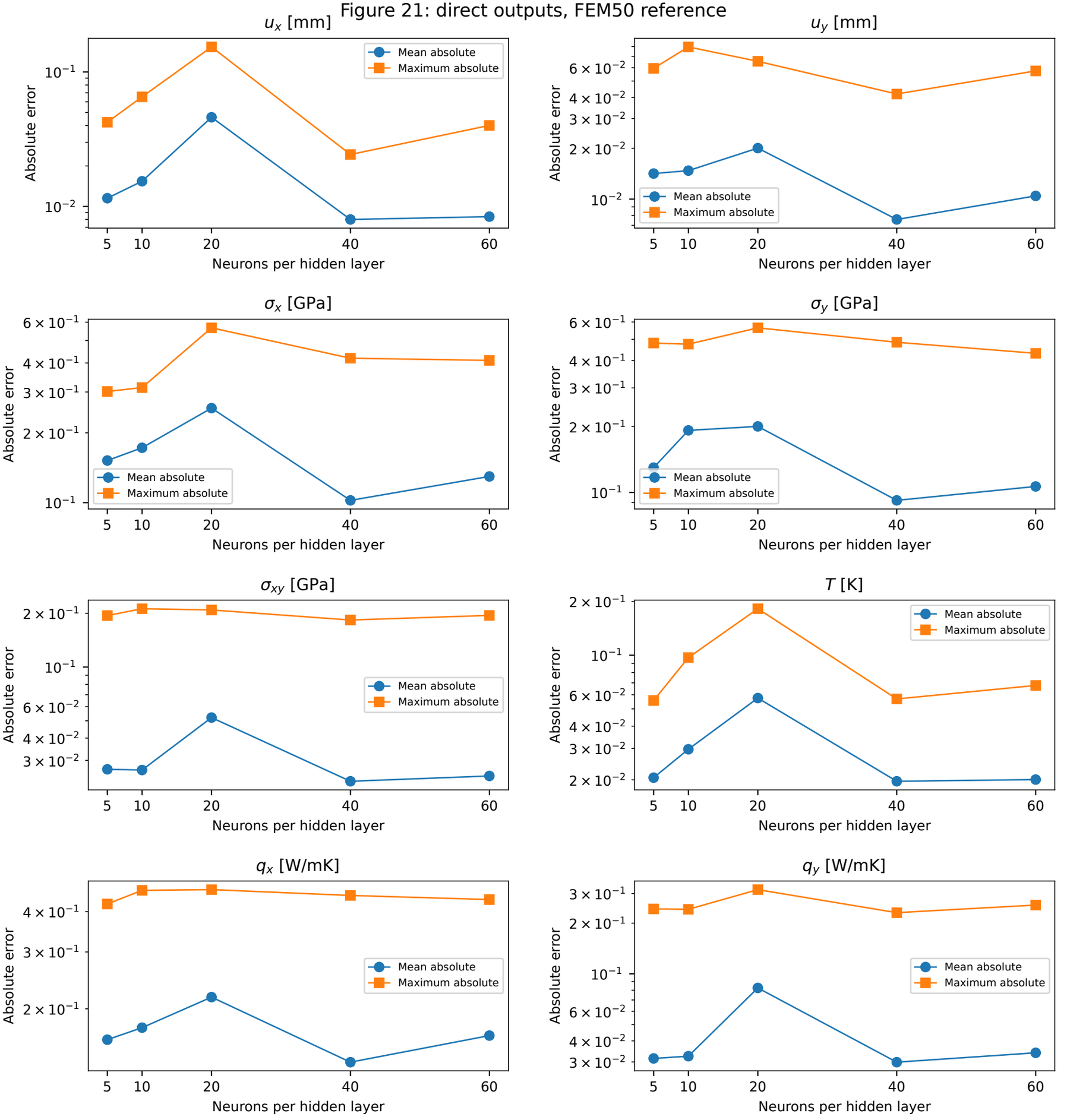
PARTIAL PROGRESS: completed jobs only | All eight fields. Maximum and mean measured errors; no assumption of monotonic improvement.

Figure 21: derived outputs, FEM50 reference



PARTIAL PROGRESS: completed jobs only | All eight fields. Maximum and mean measured errors; no assumption of monotonic improvement.

Figure 21: direct outputs, FEM50 reference



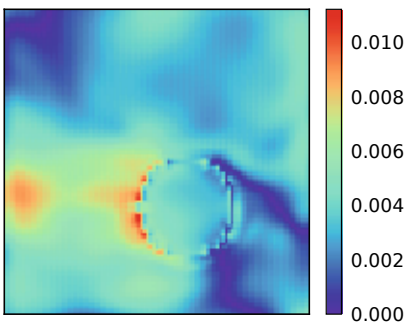
PARTIAL PROGRESS: completed jobs only | All eight fields. Maximum and mean measured errors; no assumption of monotonic improvement.

Table 4 reproduction: seconds / iteration through first100 L-BFGS updates
Rows: hidden layers; columns: neurons per layer

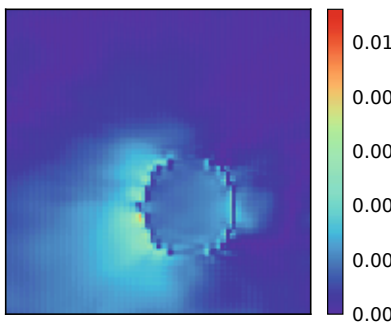
	5	10	20	40	60
1	pending	pending	pending	pending	pending
2	pending	pending	pending	pending	pending
4	pending	pending	pending	pending	pending
5	pending	pending	pending	pending	pending
6	pending	pending	pending	pending	pending

PARTIAL PROGRESS: completed jobs only | Wall time, not isolated GPU time. No completed timing jobs yet.

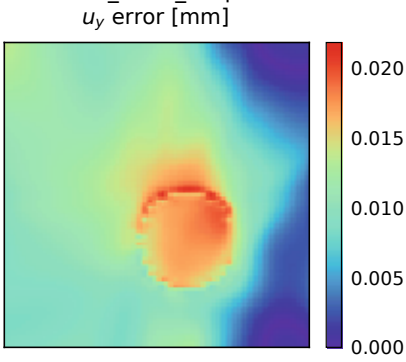
mixed_circle_coupled
u_x error [mm]



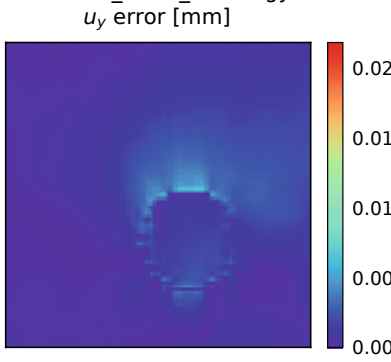
ablation_circle_noenergy
u_x error [mm]



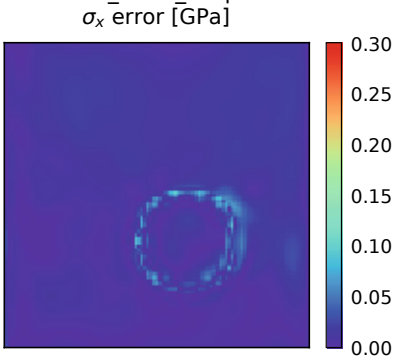
mixed_circle_coupled
u_y error [mm]



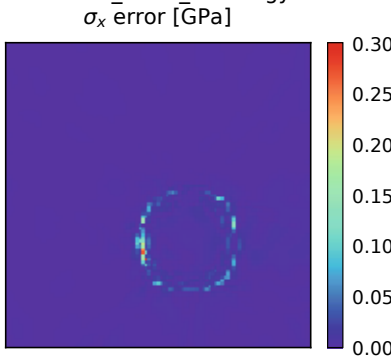
ablation_circle_noenergy
u_y error [mm]



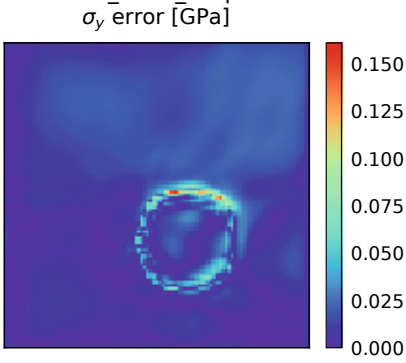
mixed_circle_coupled
σ_x error [GPa]



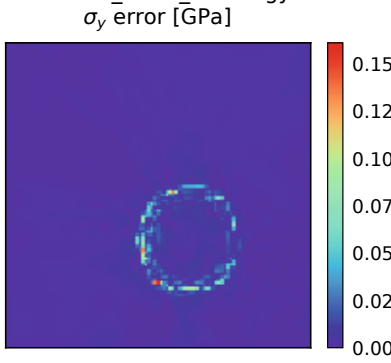
ablation_circle_noenergy
σ_x error [GPa]



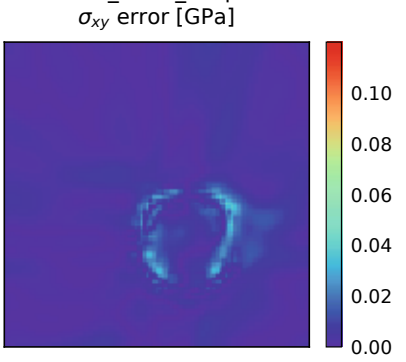
mixed_circle_coupled
σ_y error [GPa]



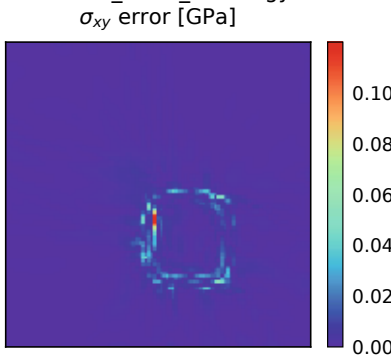
ablation_circle_noenergy
σ_y error [GPa]



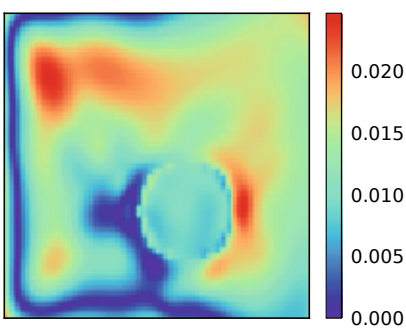
mixed_circle_coupled
σ_{xy} error [GPa]



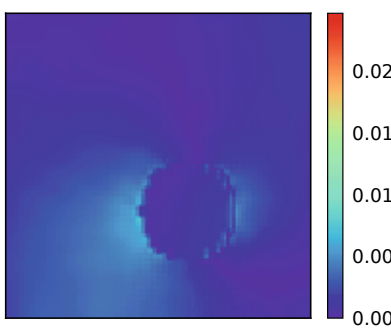
ablation_circle_noenergy
σ_{xy} error [GPa]



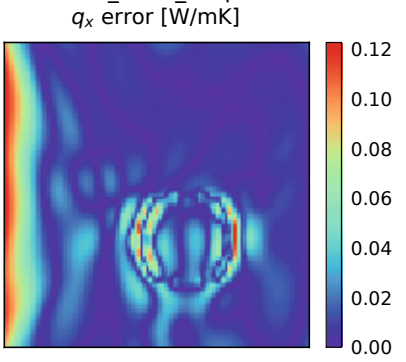
mixed_circle_coupled
T error [K]



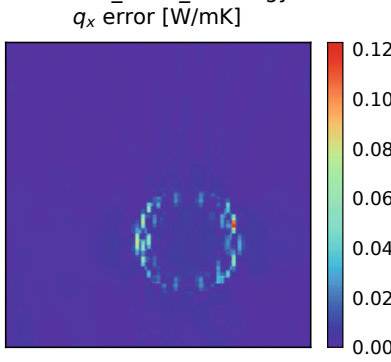
ablation_circle_noenergy
T error [K]



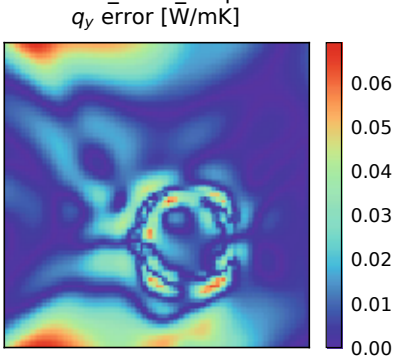
mixed_circle_coupled
q_x error [W/mK]



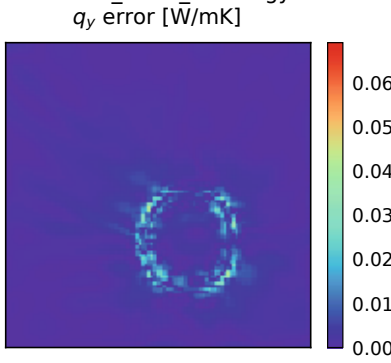
ablation_circle_noenergy
q_x error [W/mK]



mixed_circle_coupled
q_y error [W/mK]

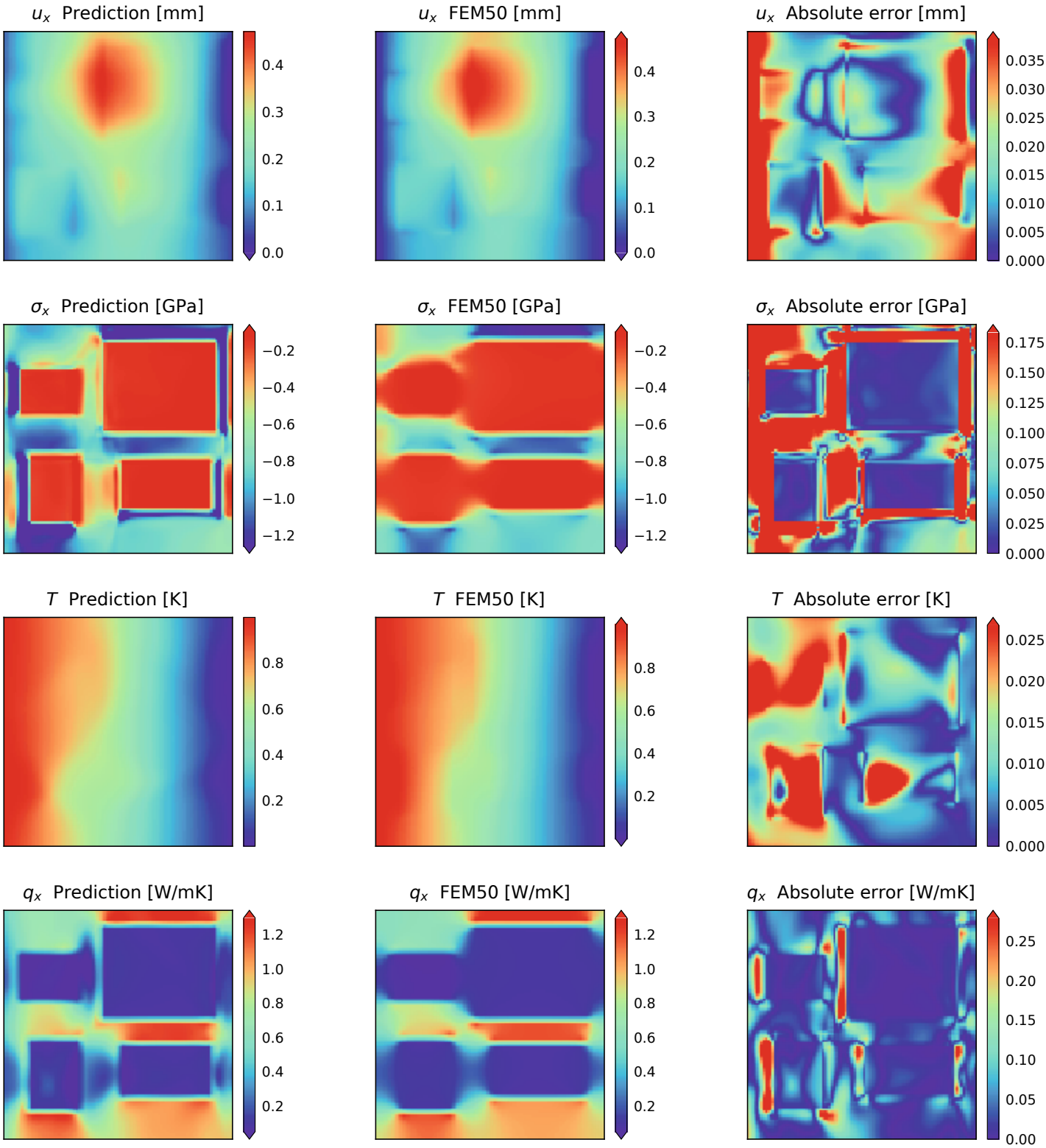


ablation_circle_noenergy
q_y error [W/mK]



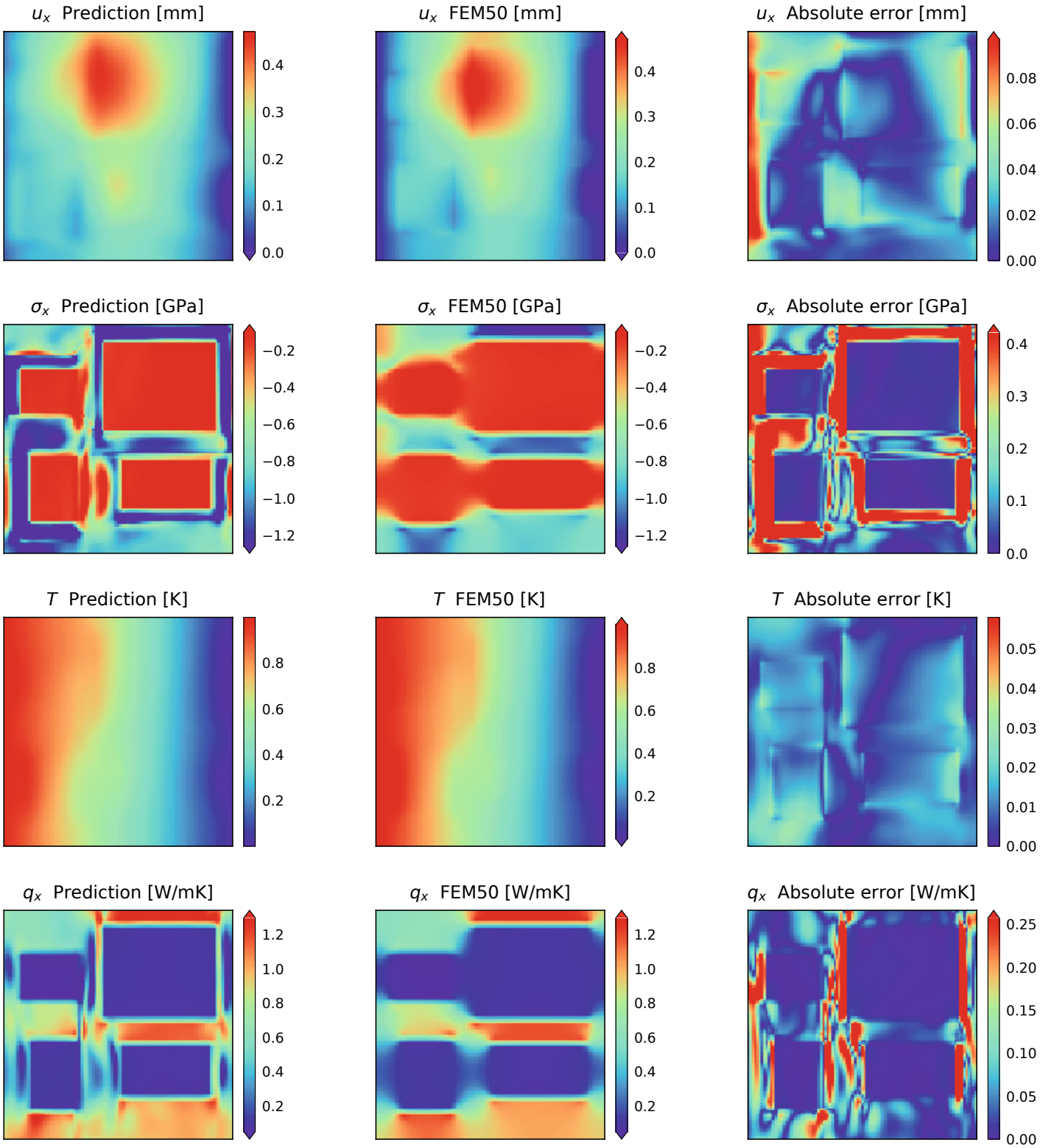
PARTIAL PROGRESS: completed jobs only | Energy omission is a reconstruction of the unquantified paper claim. Shared per-field limits across cases; derived fields.

Held-out ratio15: transfer_w0.1

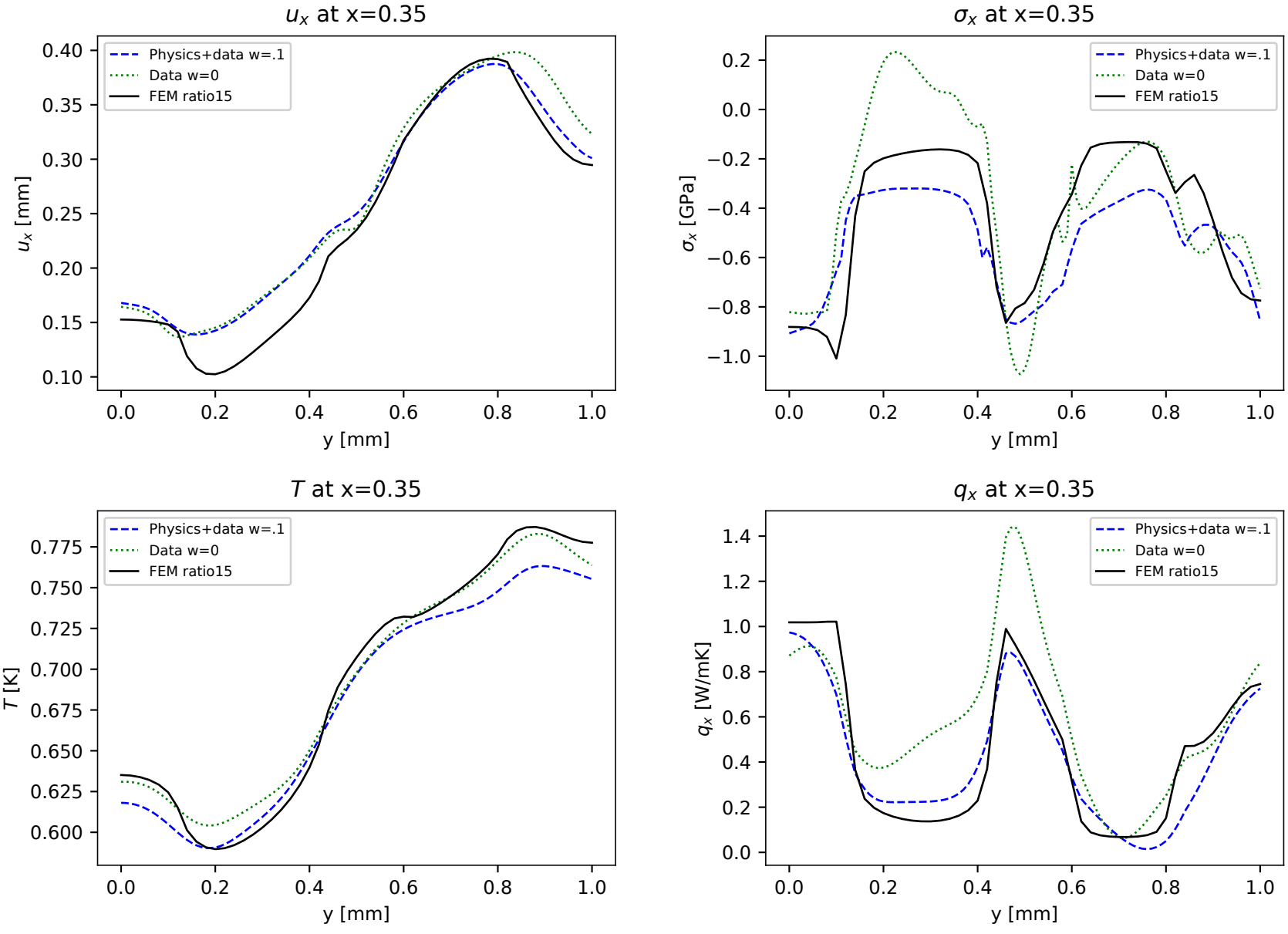


PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

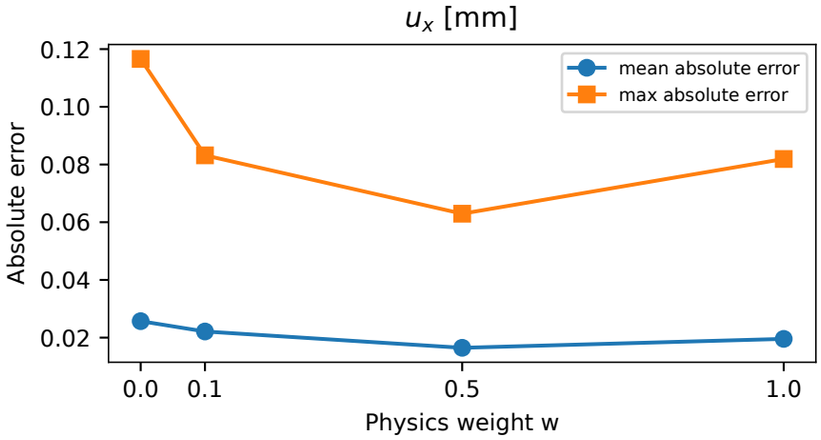
Held-out ratio15: transfer_w0

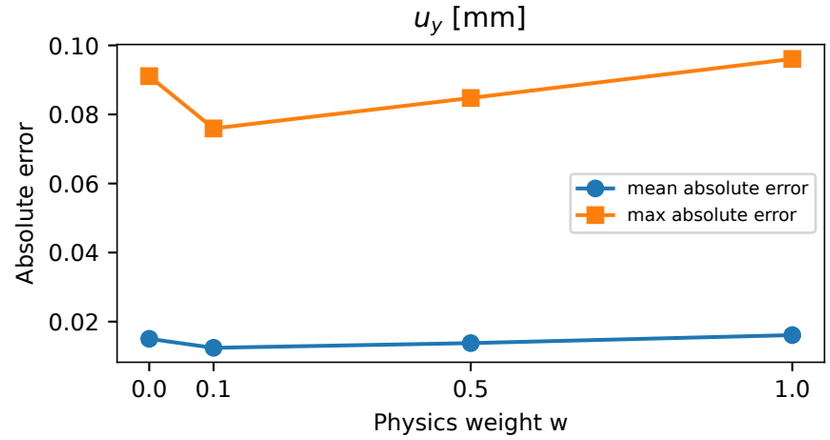


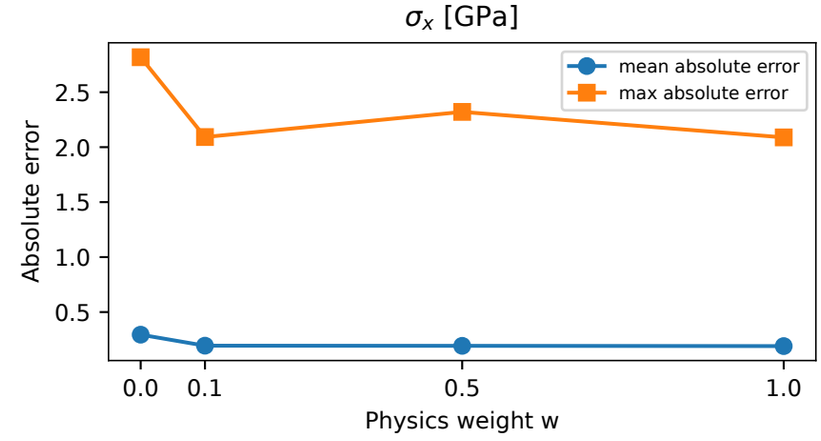
PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

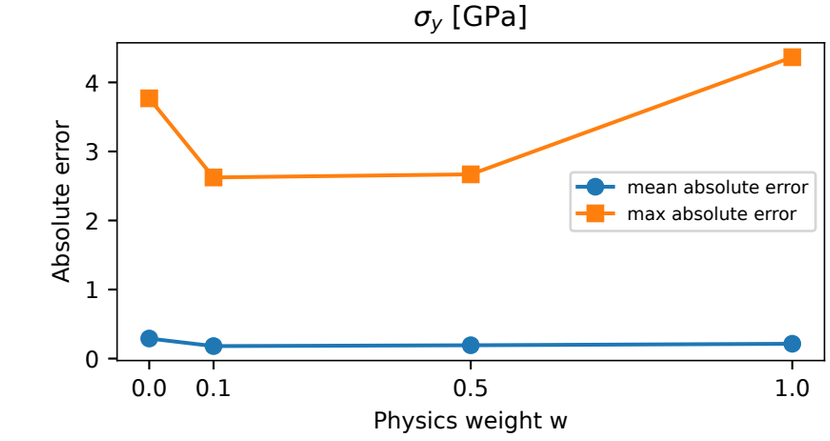


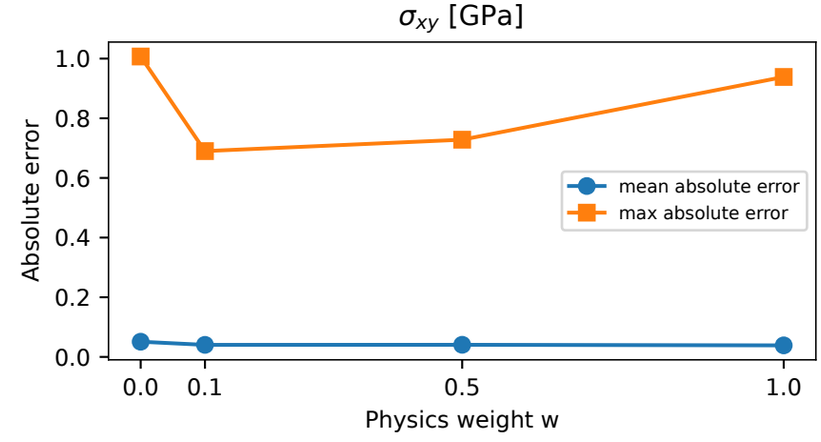
PARTIAL PROGRESS: completed jobs only | x=.35 follows panel titles and body; paper caption says.5. Held-out ratio15, no retraining.

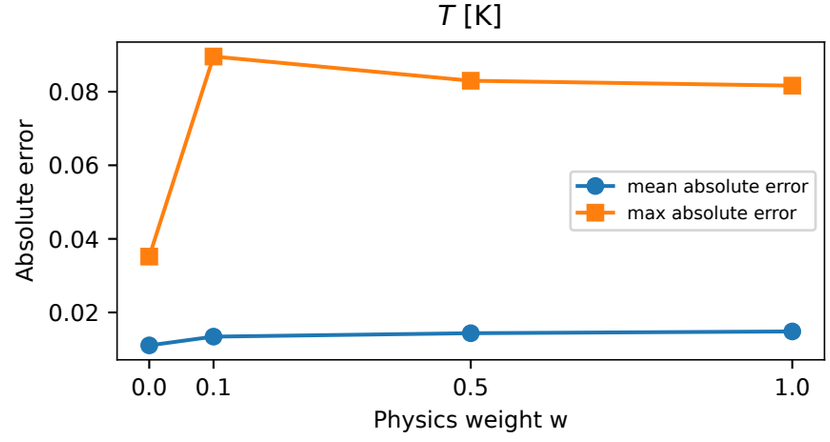


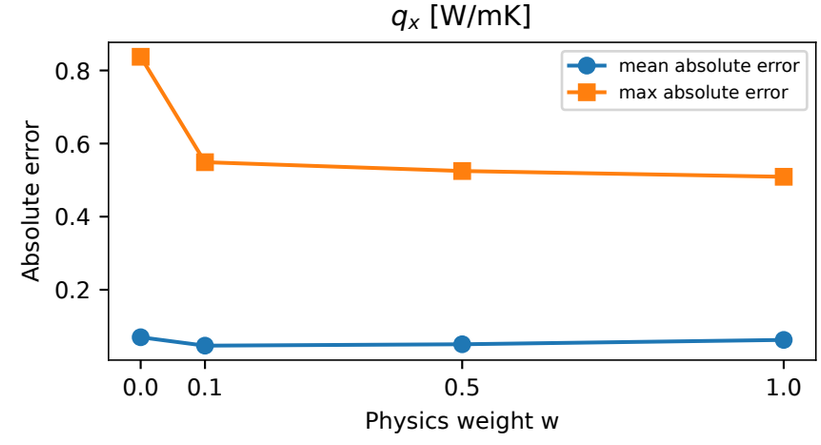


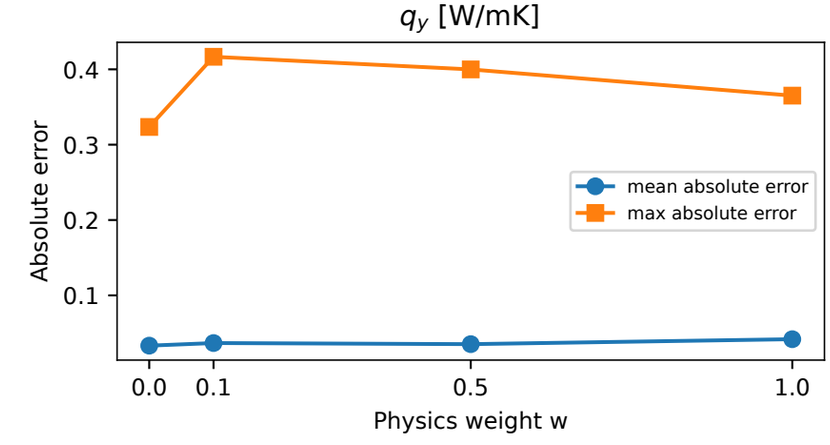






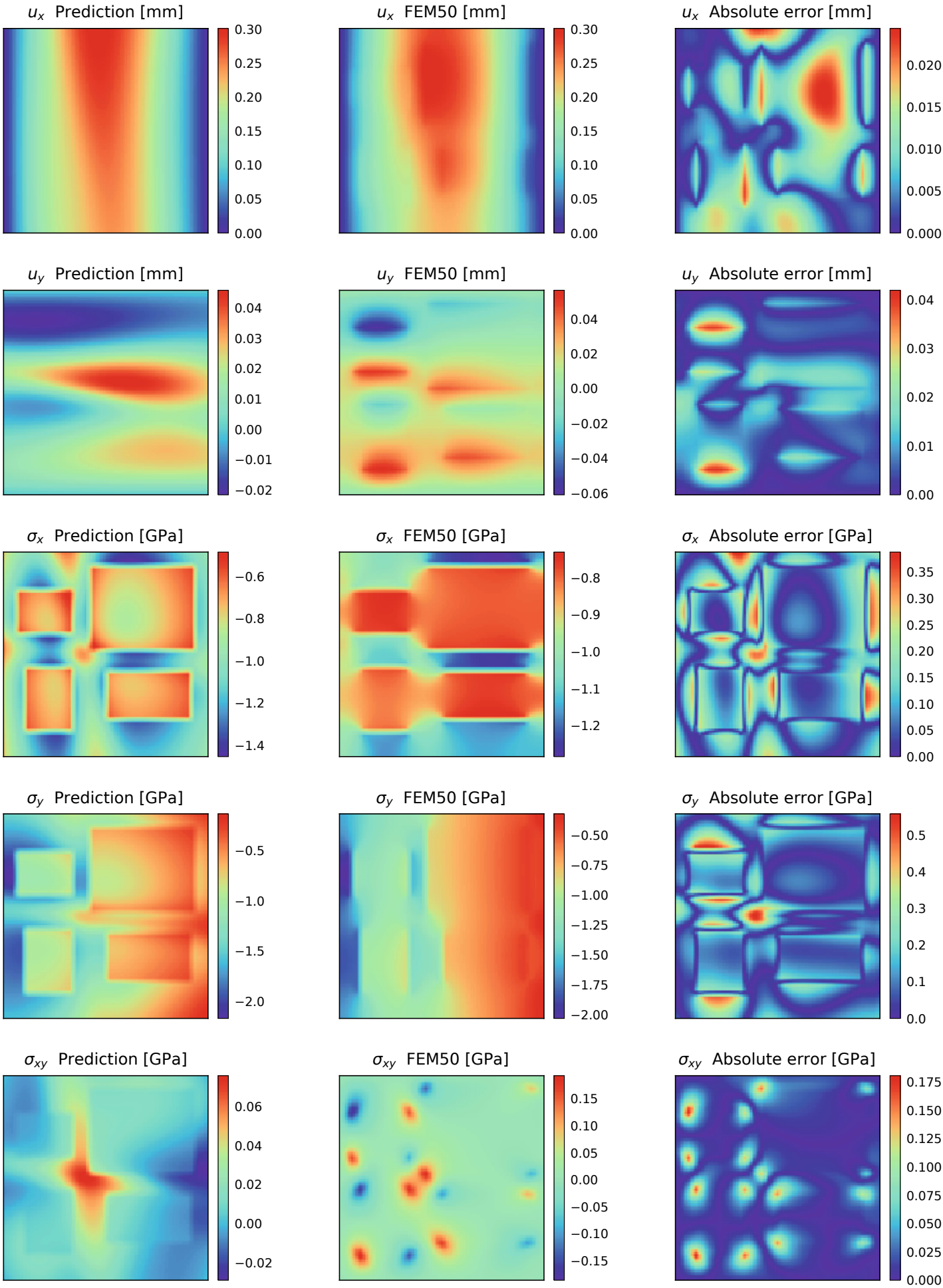




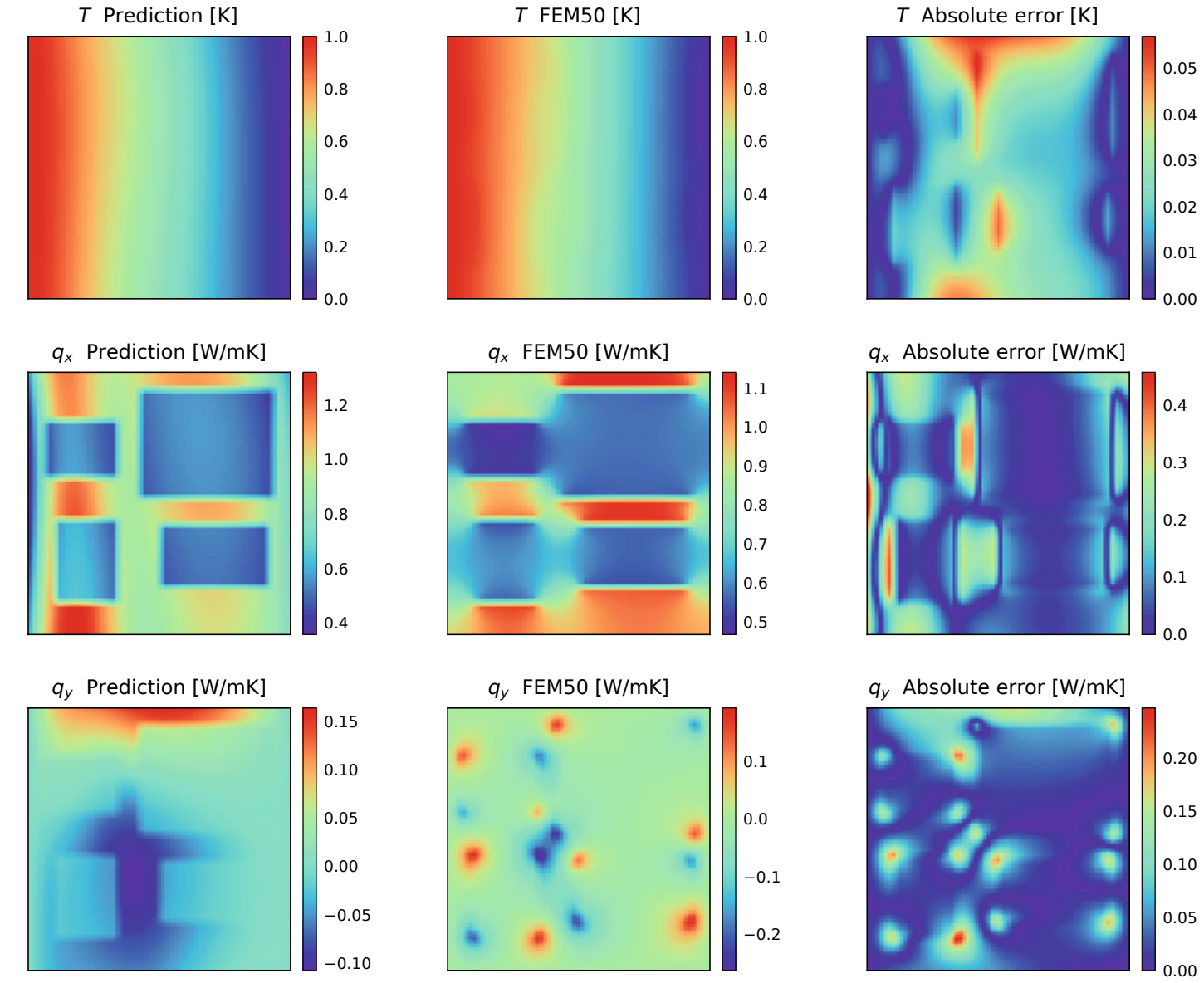


PARTIAL PROGRESS: completed jobs only | w=.5 is reconstructed. Unseen ratio15; no uniform benefit of physics assumed.

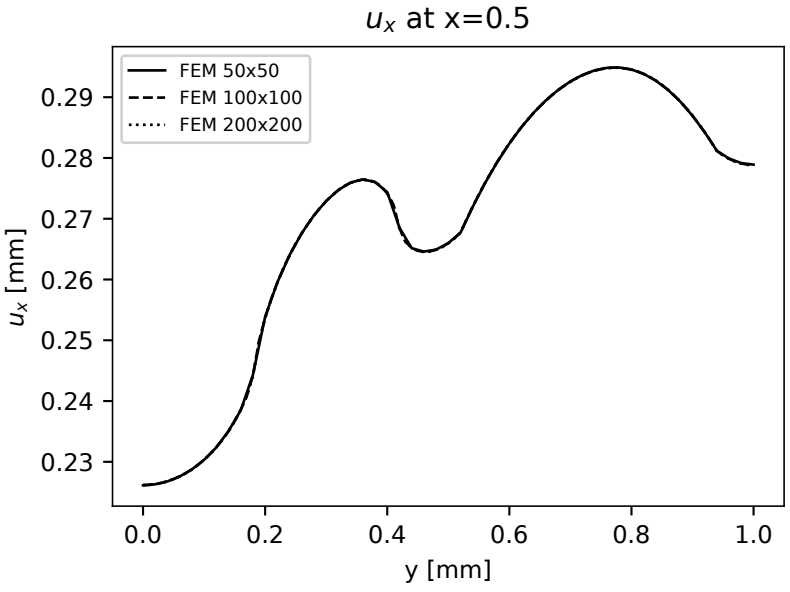
arch_l5_w40

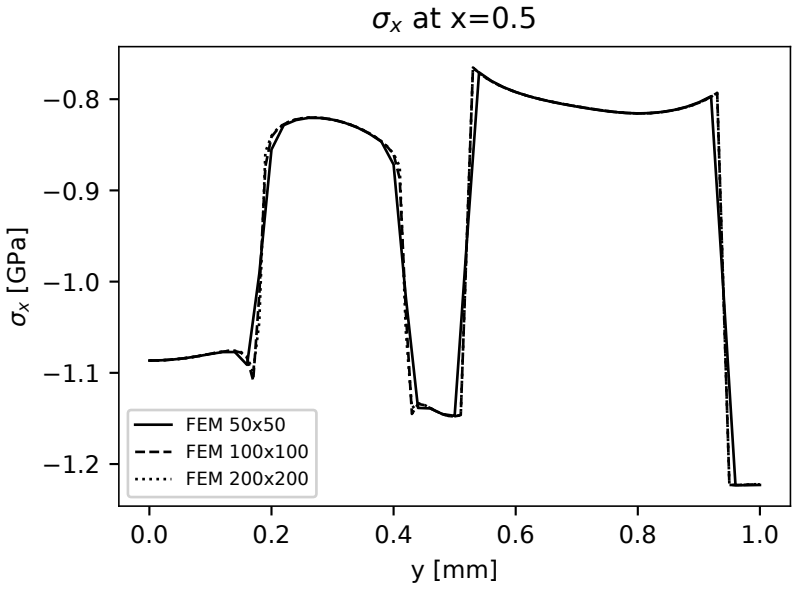


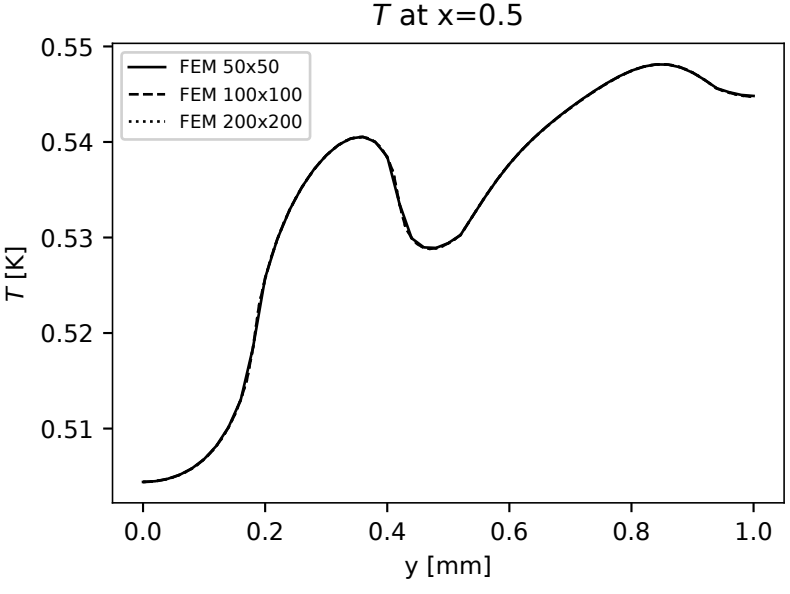
PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

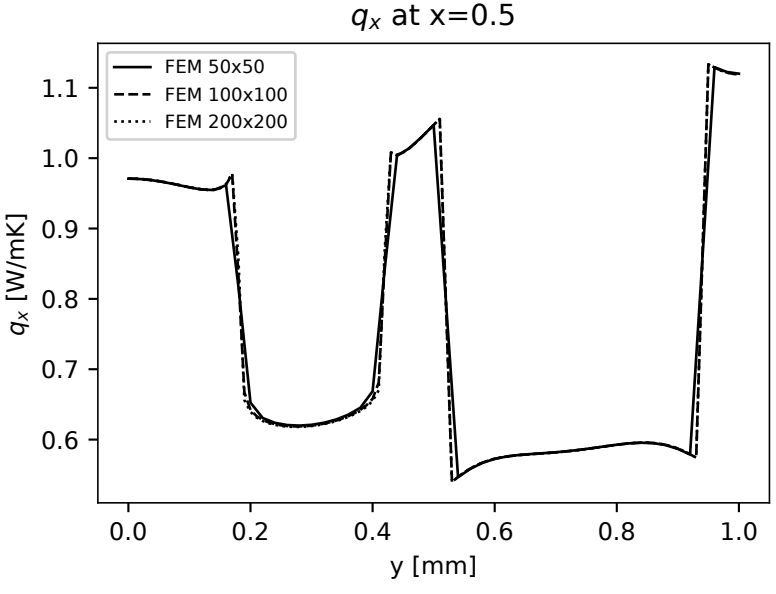


PARTIAL PROGRESS: completed jobs only | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.









PARTIAL PROGRESS: completed jobs only | Independent FE mesh convergence; quantitative norms in JSON.